

Long–Short Portfolio Return Book

Cumulative return of each factor's long–short portfolio, net of 10 bp transaction cost.
271 factors over 133 months, ordered by in-sample L–S CAGR (last 48 months, net of cost).

HOW TO READ

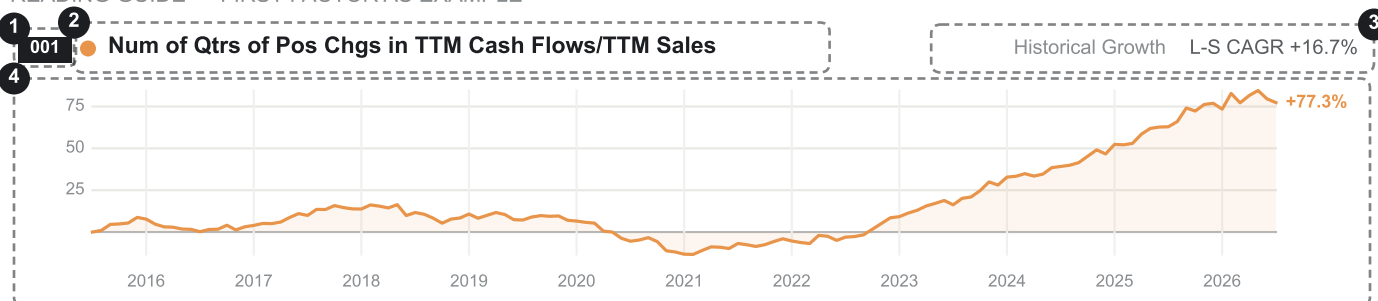
Each curve starts at 0 % in July 2015 and compounds monthly long–short returns. The line is coloured by the factor's style; the figure at the right edge is the period-end cumulative return.

FACTOR STYLES

Valuation	41	Price Momentum	54
Earnings Quality	55	Size	3
Analyst Expectations	24	Historical Growth	56
Capital Efficiency	28	Volatility	10

001 In current model portfolio (50 factors)

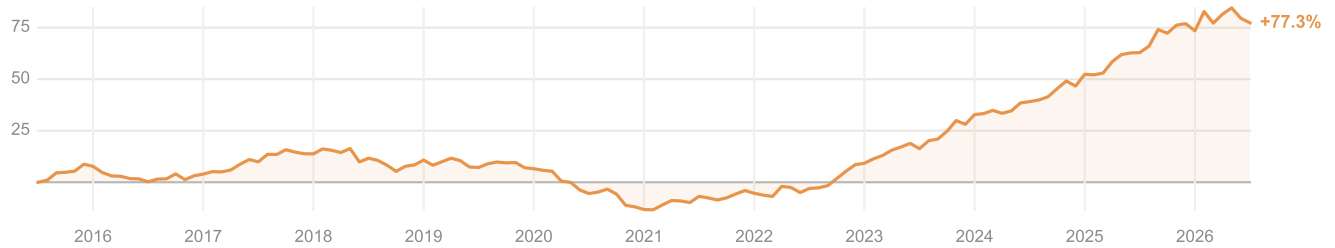
READING GUIDE — FIRST FACTOR AS EXAMPLE



001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales

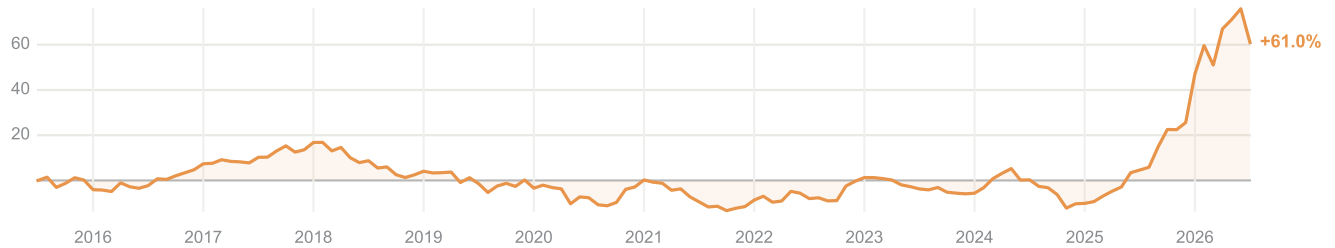
Historical Growth

CAGR +16.7%

**002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-Month Sales**

Historical Growth

CAGR +15.2%

**003 ● Pretax Income Per Share to Price**

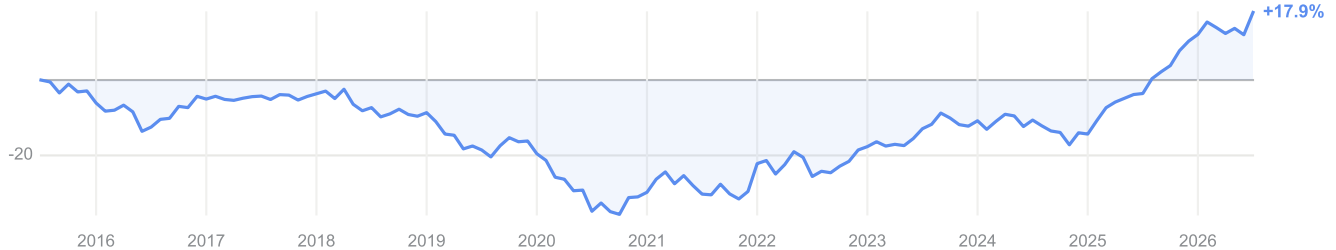
Valuation

CAGR +12.4%

**004 ● Current Liability to Price**

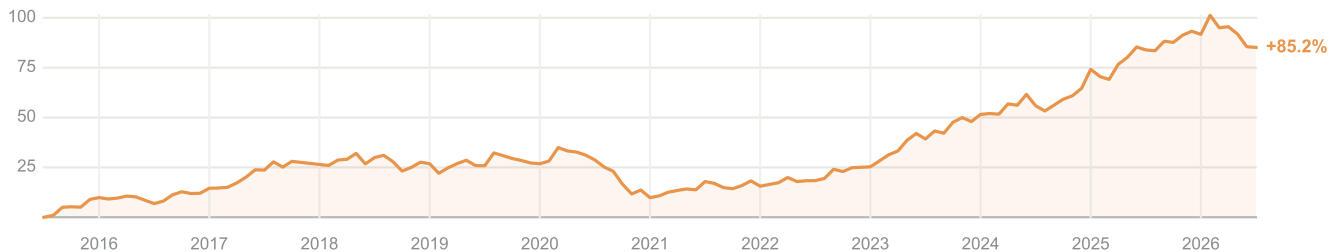
Valuation

CAGR +11.9%

**005 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flows**

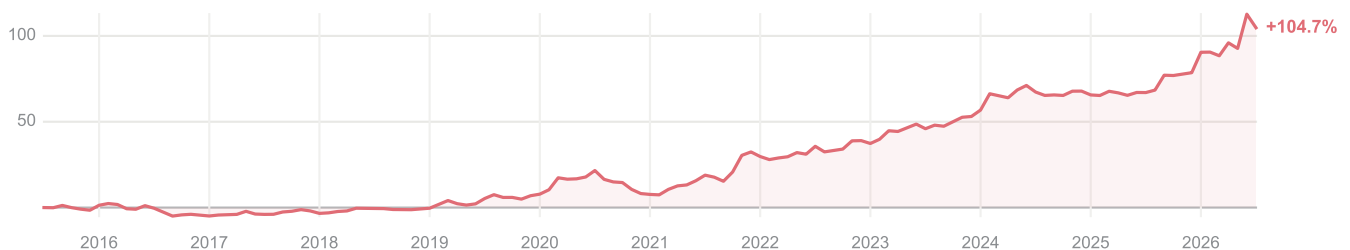
Historical Growth

CAGR +11.9%

**006 ● Operating Earnings to Assets Ratio**

Earnings Quality

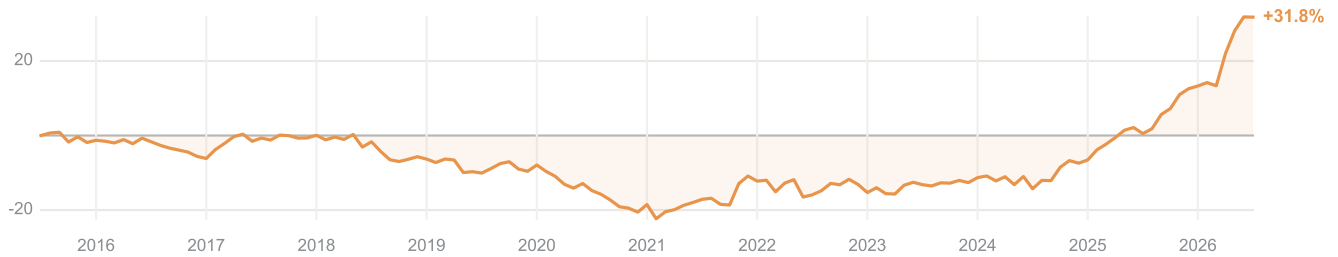
CAGR +11.8%



007 ● Change of Asset Turnover

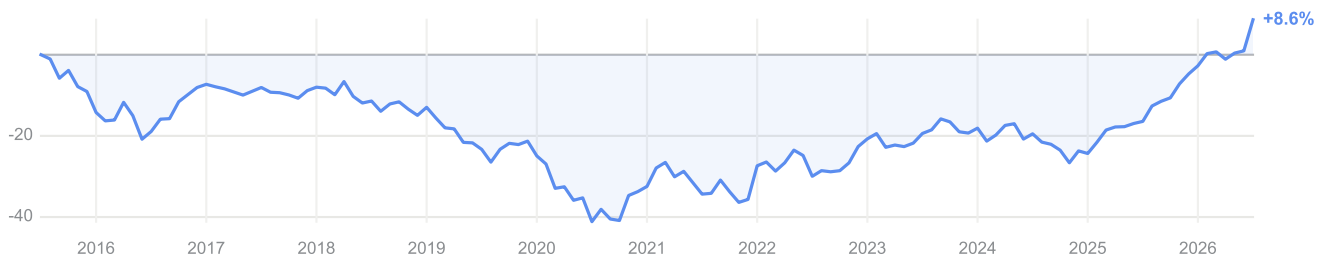
Historical Growth

CAGR +11.4%

**008 ● Assets to Price Ratio**

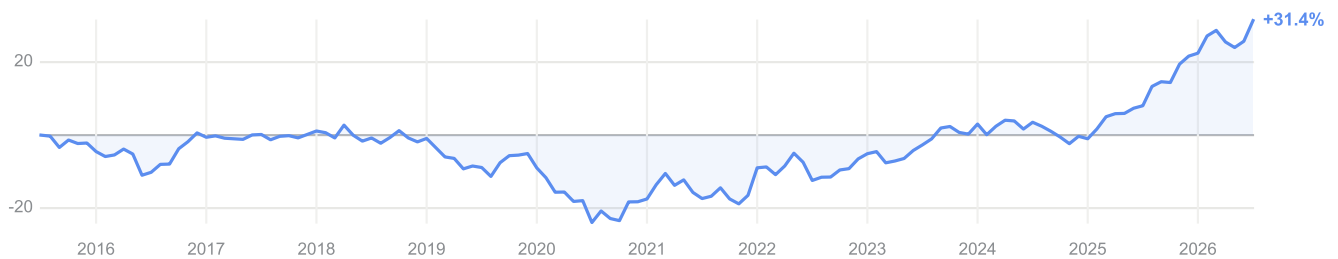
Valuation

CAGR +11.3%

**009 ● Sales to Price Ratio**

Valuation

CAGR +10.6%

**010 ● Unexpected Inventory Change**

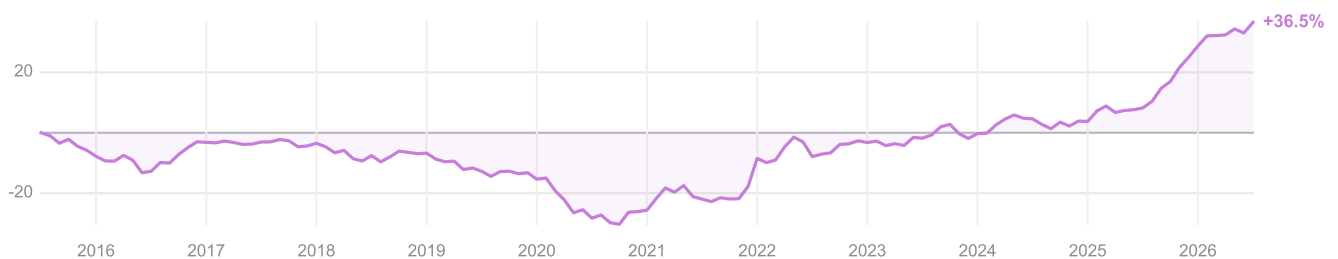
Earnings Quality

CAGR +10.5%

**011 ● Log of Total Last Quarter Assets**

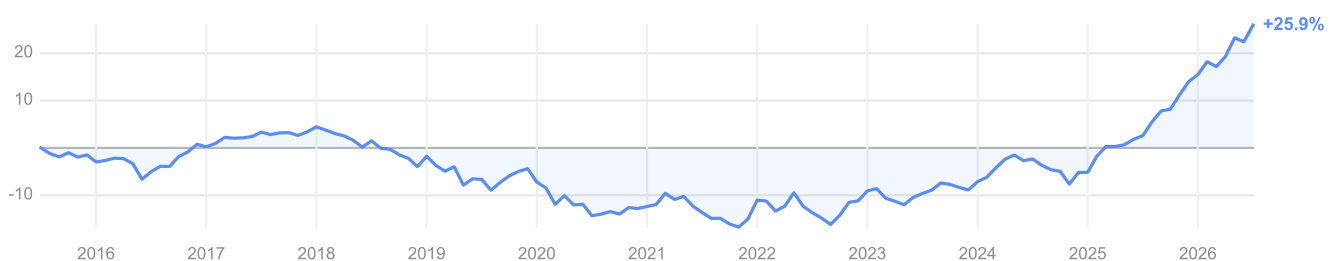
Size

CAGR +10.3%

**012 ● Inverse P/E Ratio Adj for Growth and Yield**

Valuation

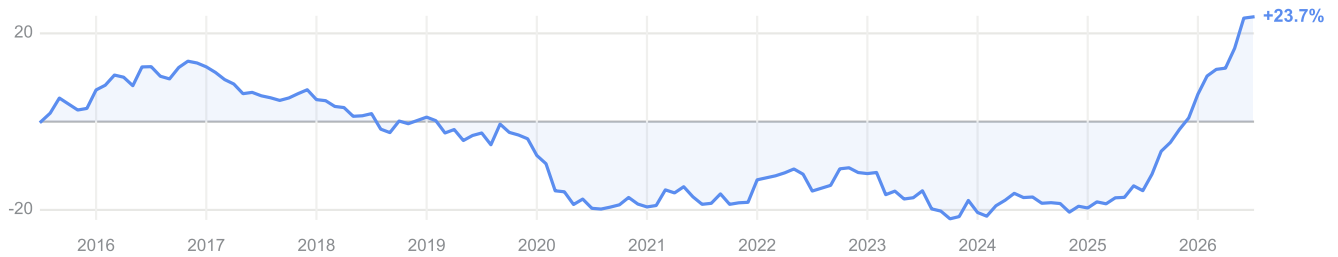
CAGR +10.3%



013 ● Indicated Annual Dividends to Price Ratio

Valuation

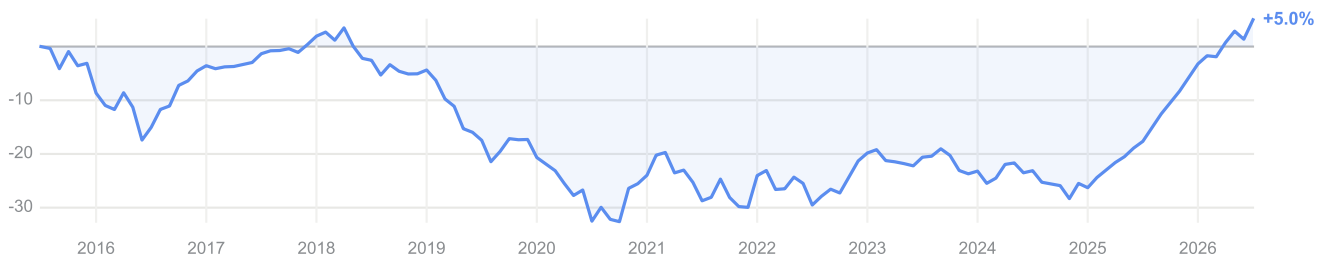
CAGR +10.1%



014 ● Cash to Price

Valuation

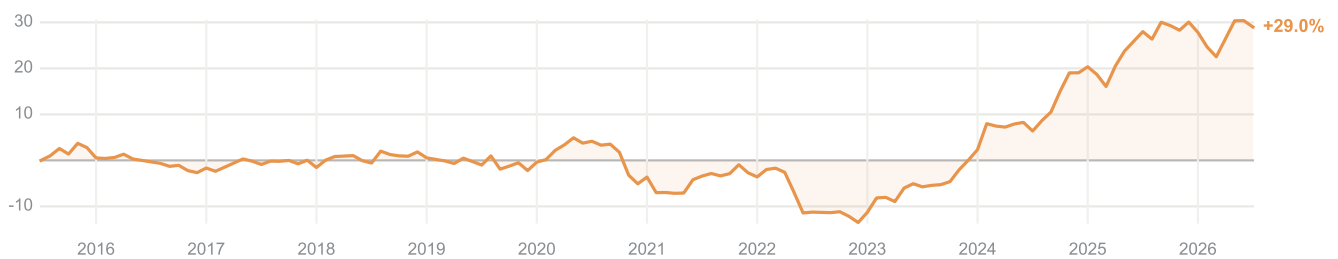
CAGR +10.0%



015 ● Adj 1Y Chg in Asset Adjusted FCF

Historical Growth

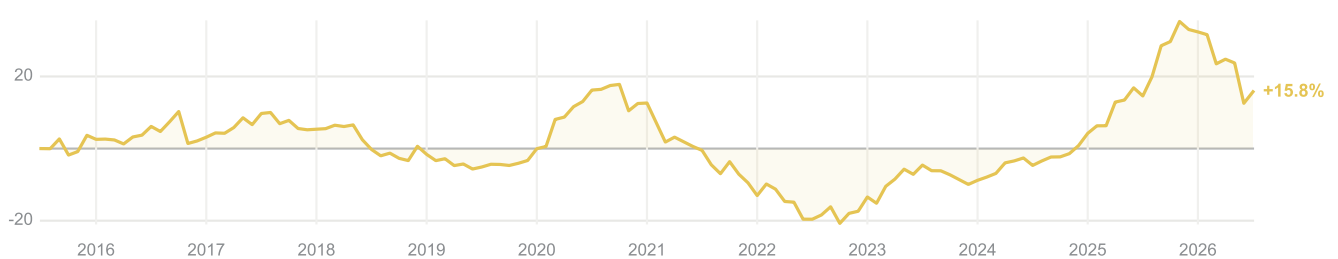
CAGR +9.8%



016 ● Long Term Debt to Equity Ratio

Capital Efficiency

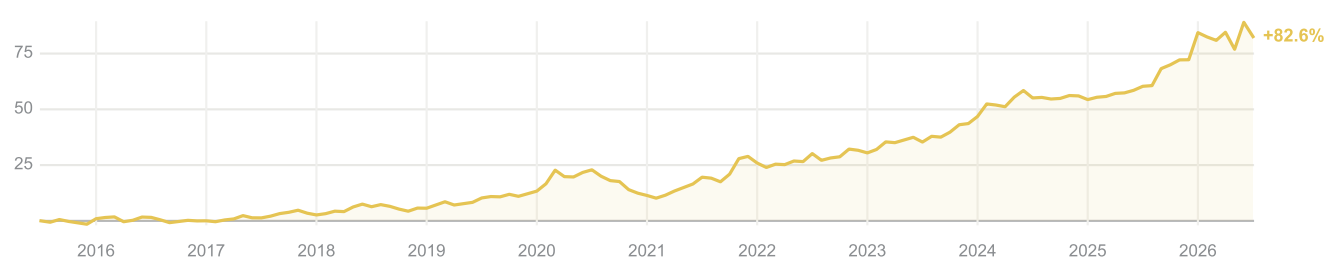
CAGR +9.8%



017 ● Return on Assets

Capital Efficiency

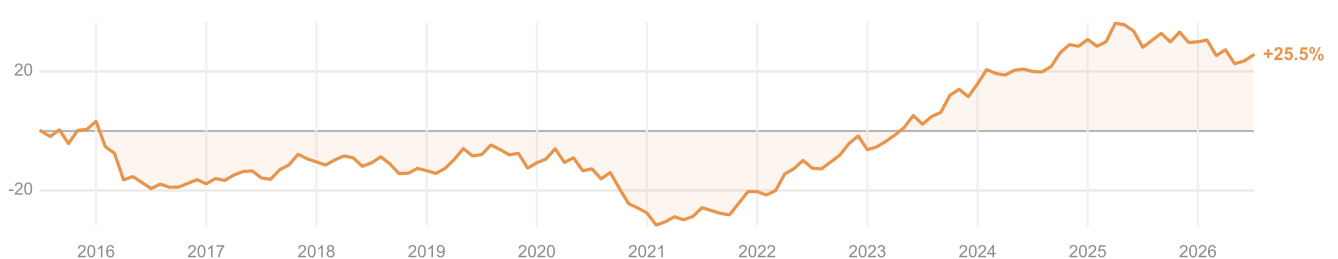
CAGR +9.8%



018 ● Number of Consecutive Qtrs of Positive Changes in TTM Quarterly Earnings

Historical Growth

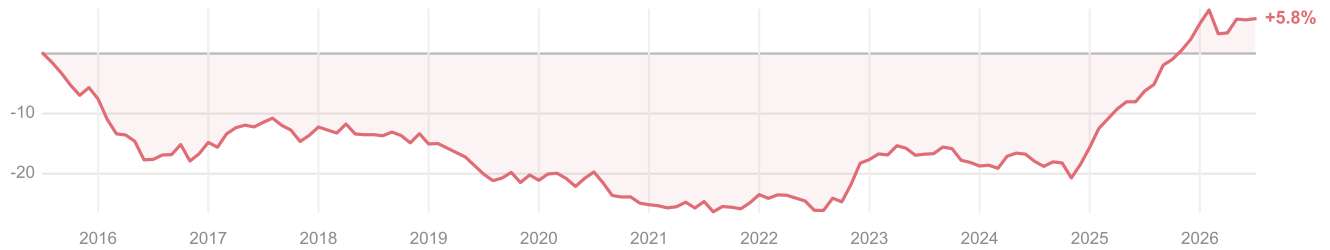
CAGR +9.7%



019 ● Gross Profit Margin

Earnings Quality

CAGR +9.7%

**020 ● Adjusted 3M Revision in FY1 EPS Estimate**

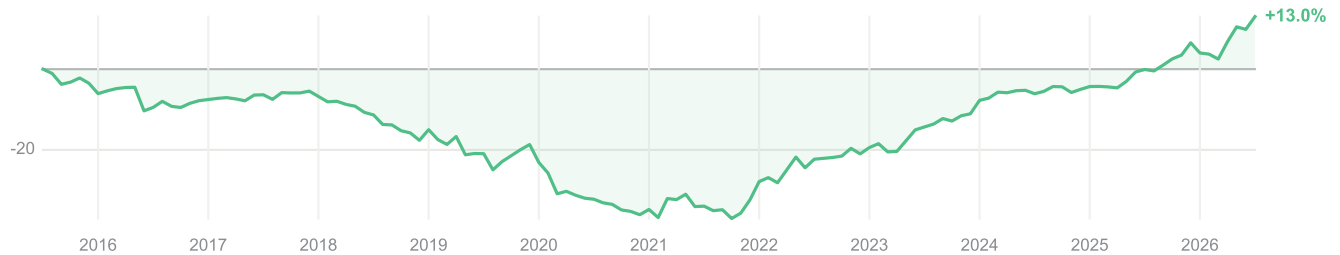
Analyst Expectations

CAGR +9.7%

**021 ● Adjusted Inverse PEG**

Analyst Expectations

CAGR +9.7%

**022 ● Analyst Dispersion for FY2 EPS**

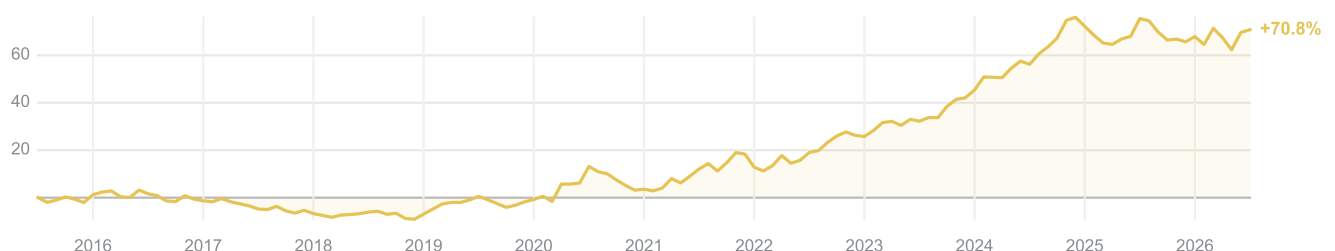
Analyst Expectations

CAGR +9.6%

**023 ● Return on Equity**

Capital Efficiency

CAGR +9.5%

**024 ● Forward Earnings to Price**

Valuation

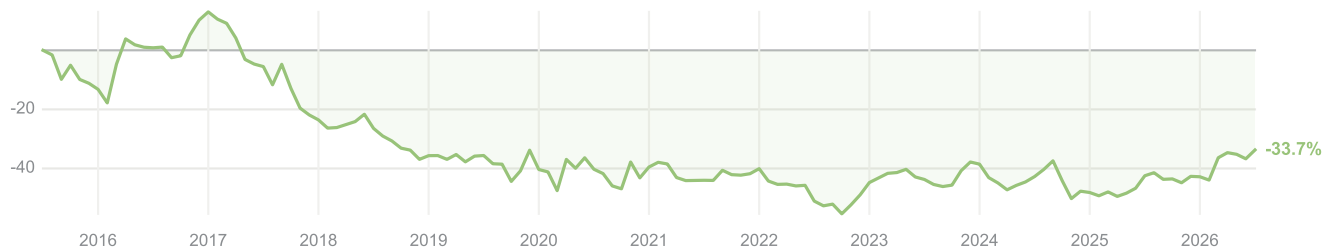
CAGR +9.5%



025 ● **Sharpe Ratio**

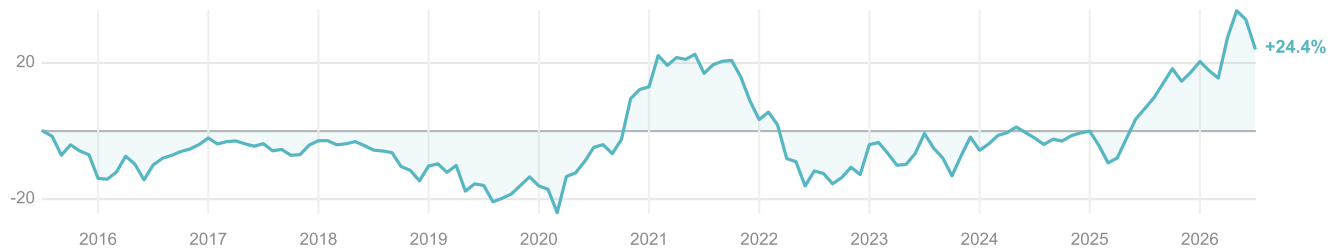
Price Momentum

CAGR +9.4%

026 ● **12M Realized Price Volatility**

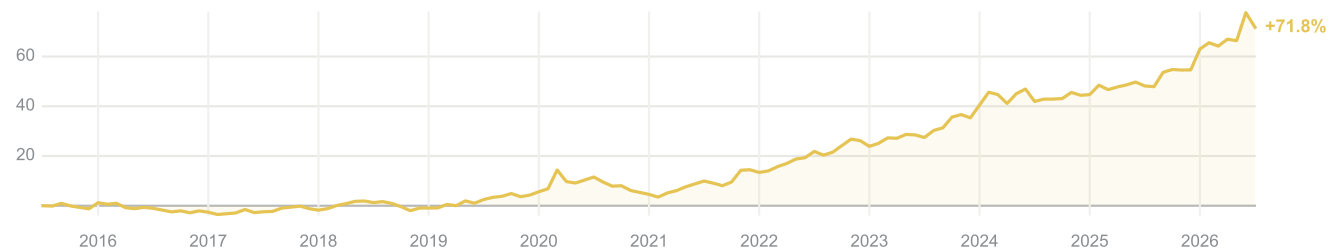
Volatility

CAGR +9.4%

027 ● **Return on Invested Capital**

Capital Efficiency

CAGR +9.4%

028 ● **Quick Ratio**

Earnings Quality

CAGR +9.3%

029 ● **3Y Chg in Operating Cash Flow to Price**

Historical Growth

CAGR +9.2%

030 ● **Operating Earnings to Price Ratio**

Valuation

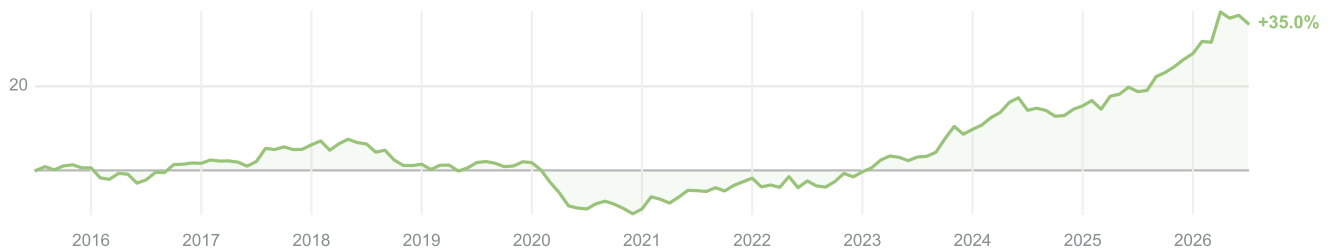
CAGR +9.1%



031 ● Short Interest Trading Volume

Price Momentum

CAGR +9.0%

**032 ● Trailing 12-Month Earnings/Trailing 12-Month Sales**

Earnings Quality

CAGR +9.0%

**033 ● Tobin's Q Ratio**

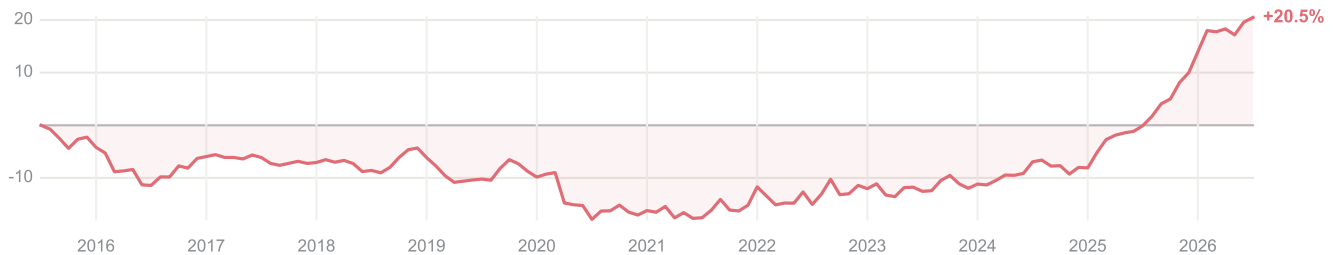
Capital Efficiency

CAGR +8.9%

**034 ● ROA 20 Qtr Standard Deviation**

Earnings Quality

CAGR +8.8%

**035 ● Adj Growth Flow to Price**

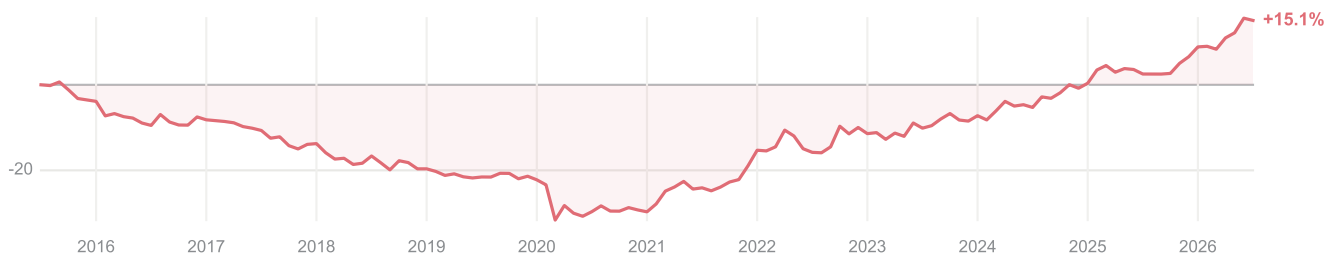
Valuation

CAGR +8.7%

**036 ● Working Capital to Sales**

Earnings Quality

CAGR +8.4%



037 ● 3M Revision in FY1 EPS Estimate

Analyst Expectations

CAGR +8.3%

**038 ● Adjusted Forward Earnings to Price**

Valuation

CAGR +8.2%

**039 ● Normalized EP**

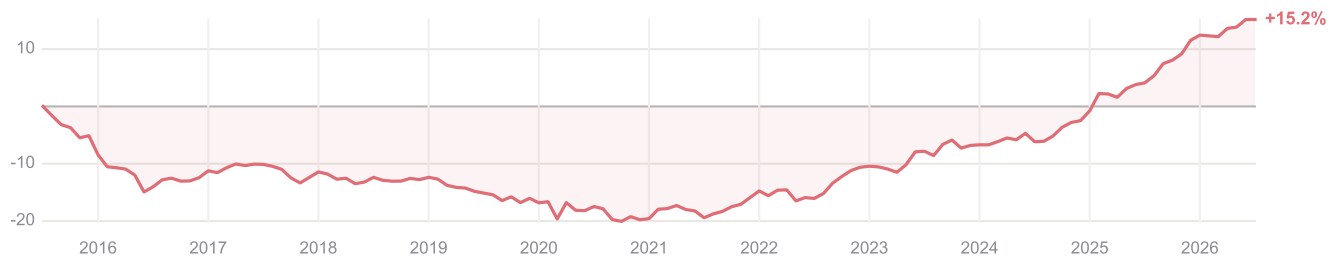
Valuation

CAGR +8.2%

**040 ● Balance Sheet Accruals**

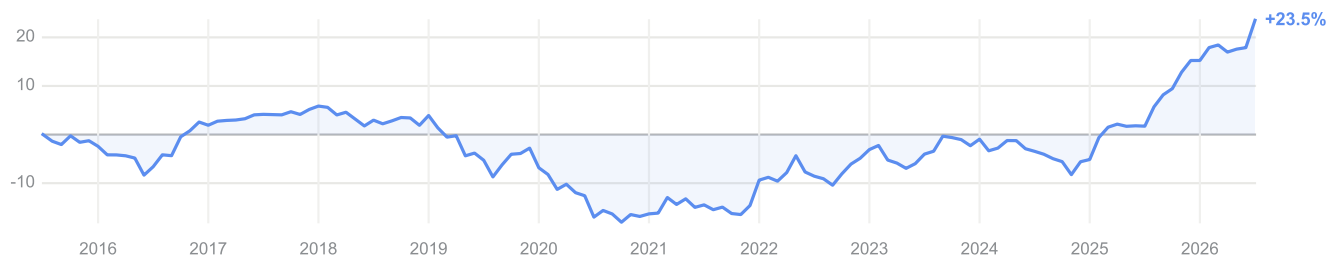
Earnings Quality

CAGR +8.1%

**041 ● Earnings to Price**

Valuation

CAGR +8.1%

**042 ● Slope of 52W Price Trend**

Price Momentum

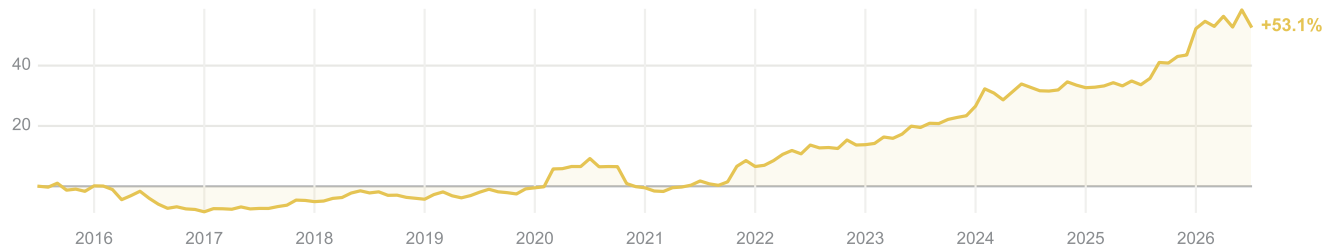
CAGR +8.1%



043 ● EBITDA to Assets

Capital Efficiency

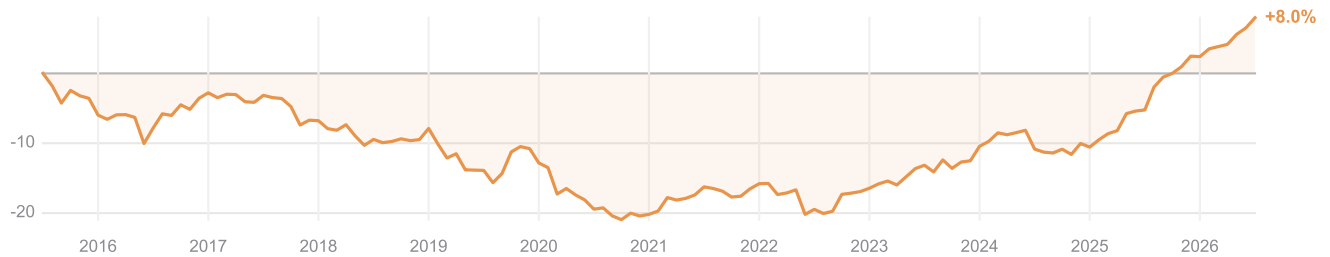
CAGR +8.0%



044 ● 3Y Chg in Cash Flow to Price

Historical Growth

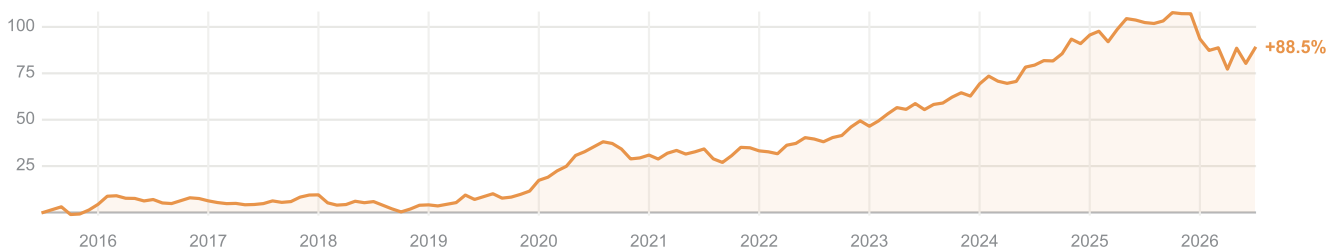
CAGR +8.0%



045 ● Consecutive Quarters of Declines in TTM Overhead/Sales

Historical Growth

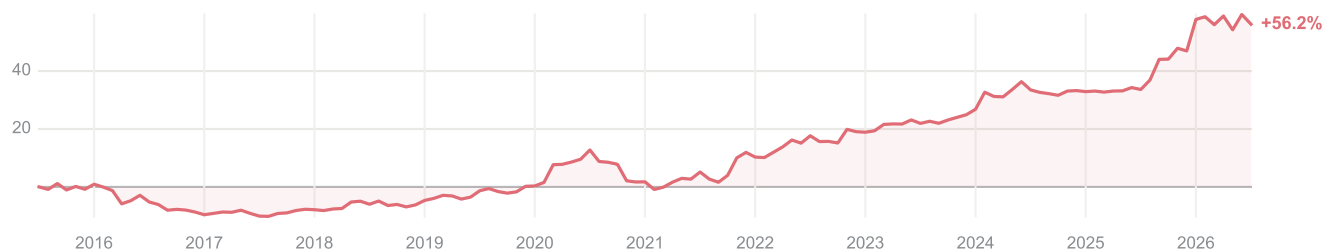
CAGR +8.0%



046 ● Cash Flow to Total Asset

Earnings Quality

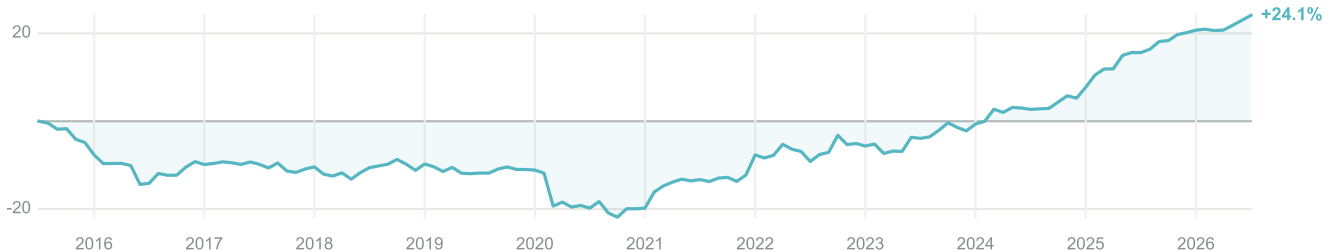
CAGR +8.0%



047 ● Book Leverage

Volatility

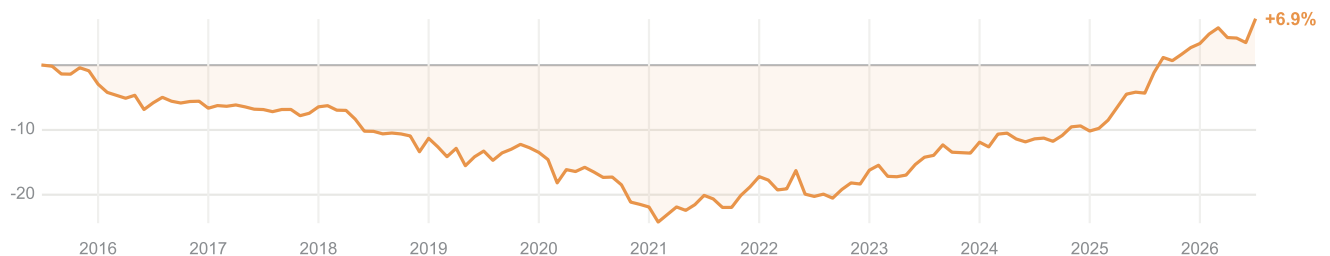
CAGR +7.9%

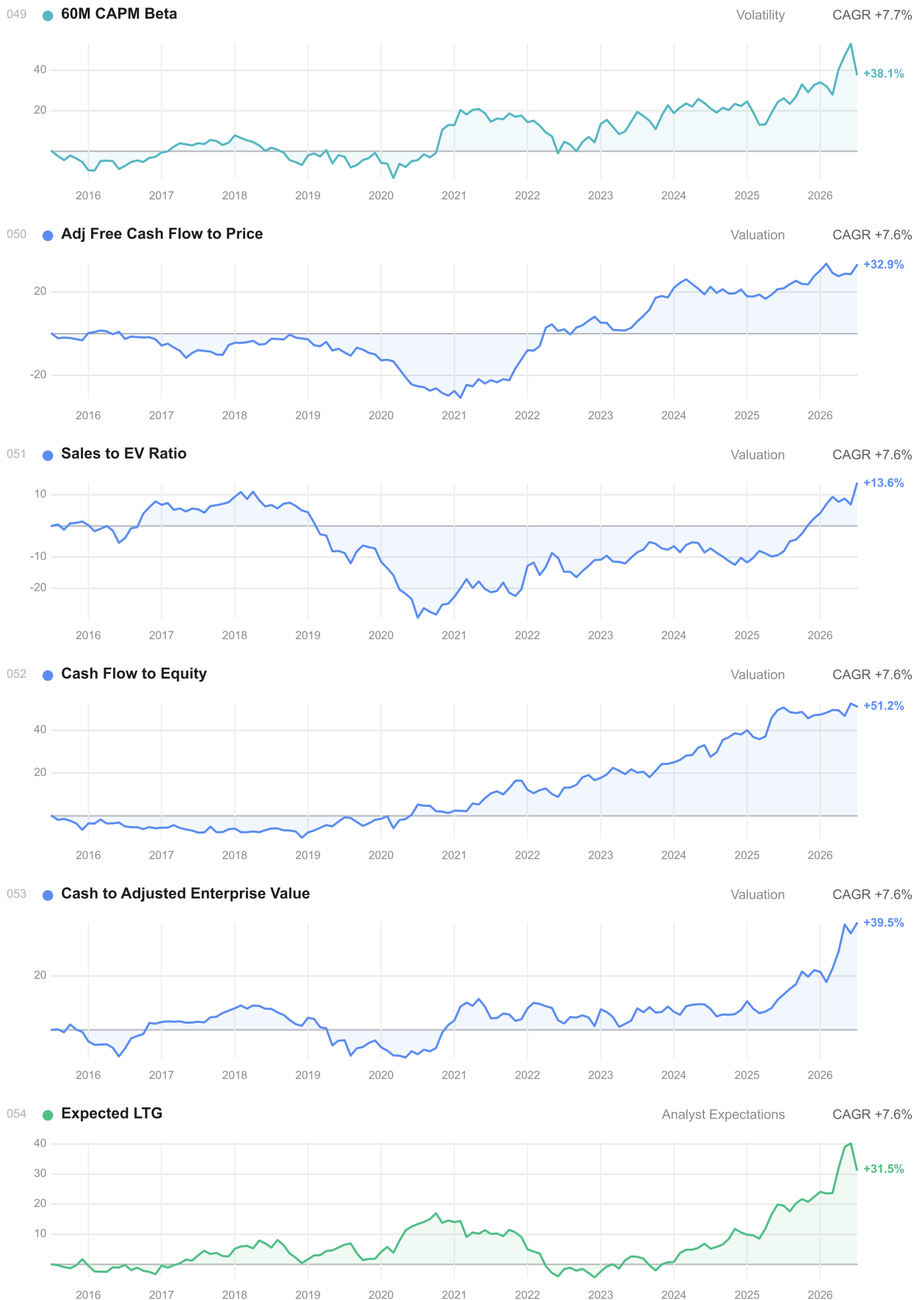


048 ● 1Y Chg in Sales to Price

Historical Growth

CAGR +7.8%

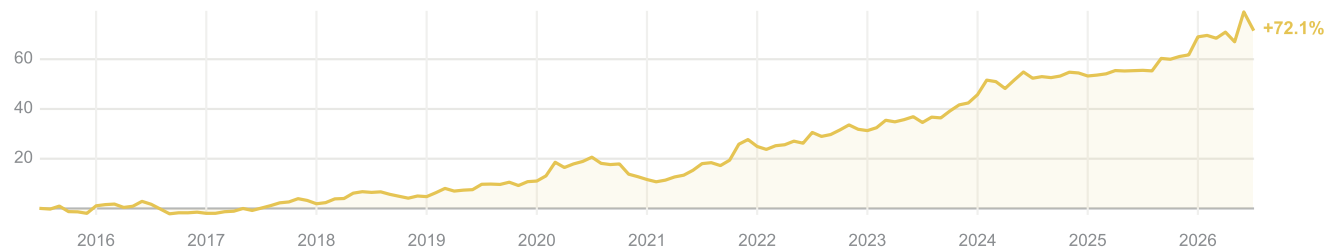




055 ● **EBIT to Assets**

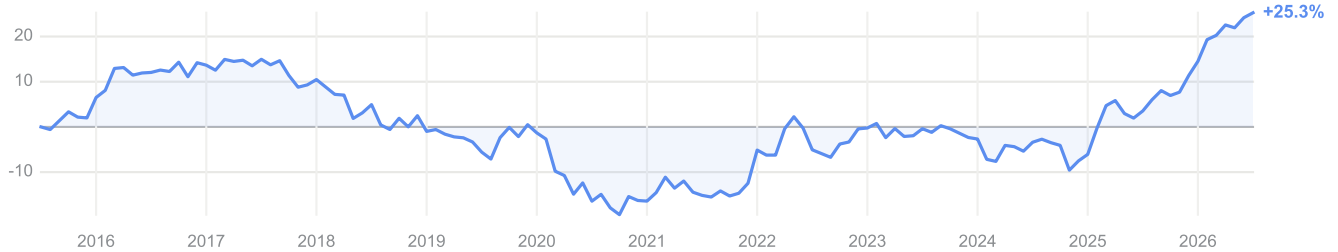
Capital Efficiency

CAGR +7.6%

056 ● **Dividends to Price Ratio**

Valuation

CAGR +7.6%

057 ● **Short Interest Ratio**

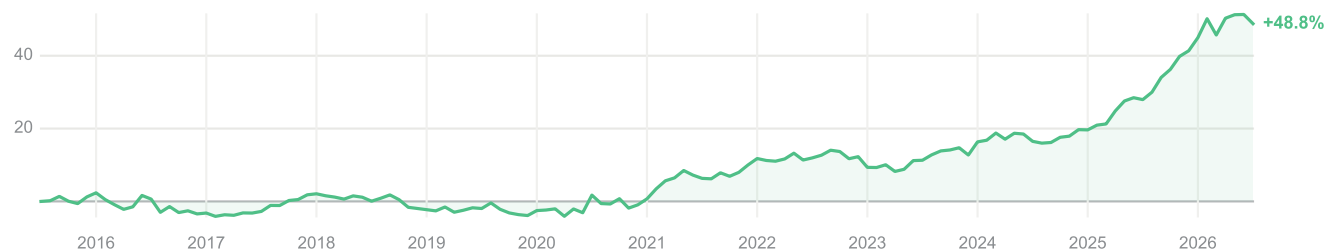
Price Momentum

CAGR +7.5%

058 ● **3M Revision in FY2 EPS Estimate**

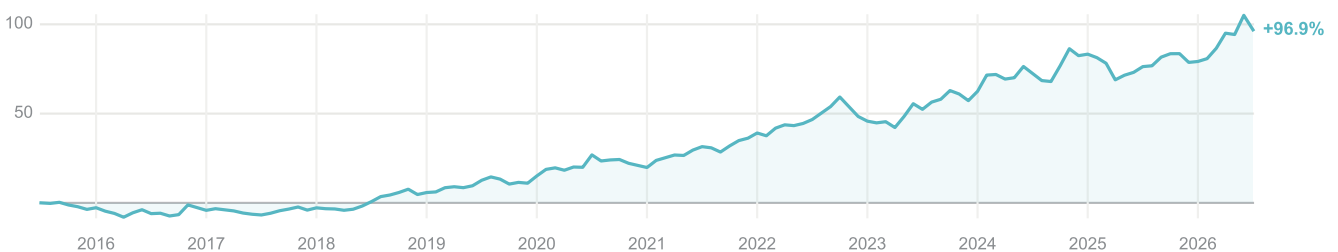
Analyst Expectations

CAGR +7.4%

059 ● **Amihud Illiquidity Measure**

Volatility

CAGR +7.3%

060 ● **Inverse PEG**

Analyst Expectations

CAGR +7.2%



061 ● Adjusted EBIT to Price

Valuation

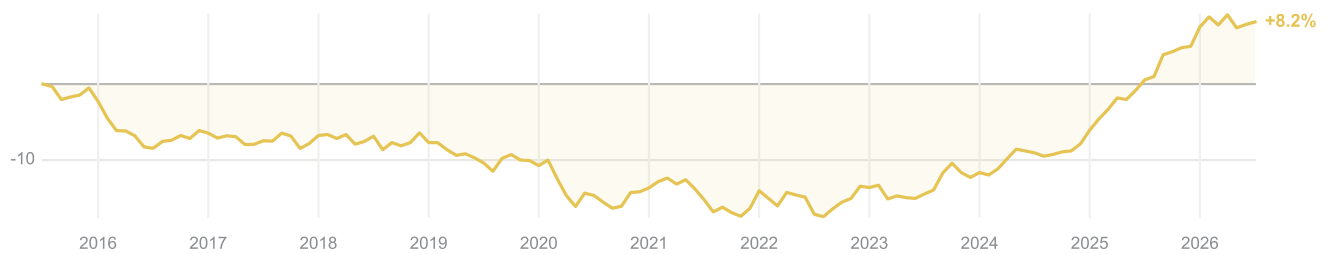
CAGR +7.1%



062 ● Capital Expenditure to Total Assets

Capital Efficiency

CAGR +7.1%



063 ● Net Cash Flow to Price

Valuation

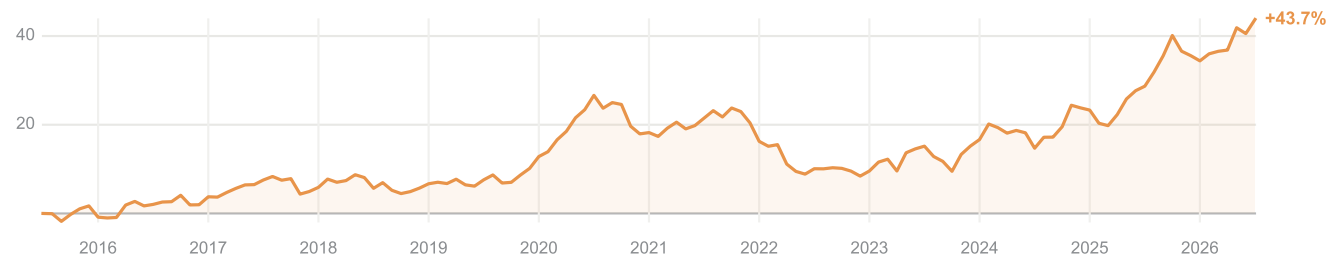
CAGR +7.1%



064 ● 3Y Chg in Operating Cash Flow per Share

Historical Growth

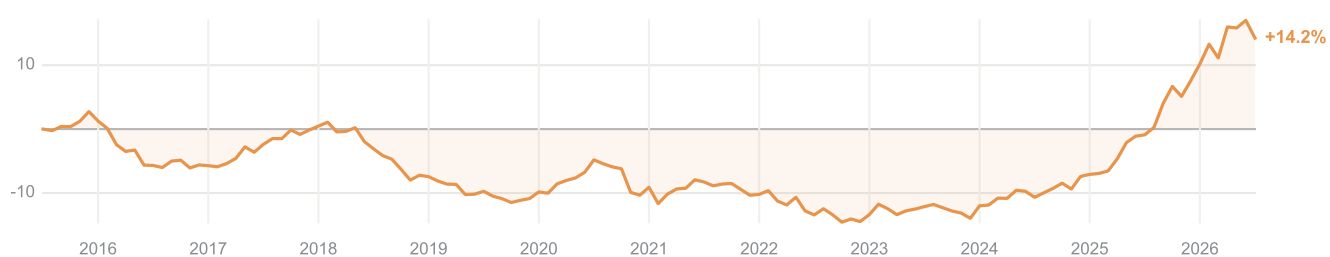
CAGR +7.1%



065 ● 1Y Chg in Operating Margin

Historical Growth

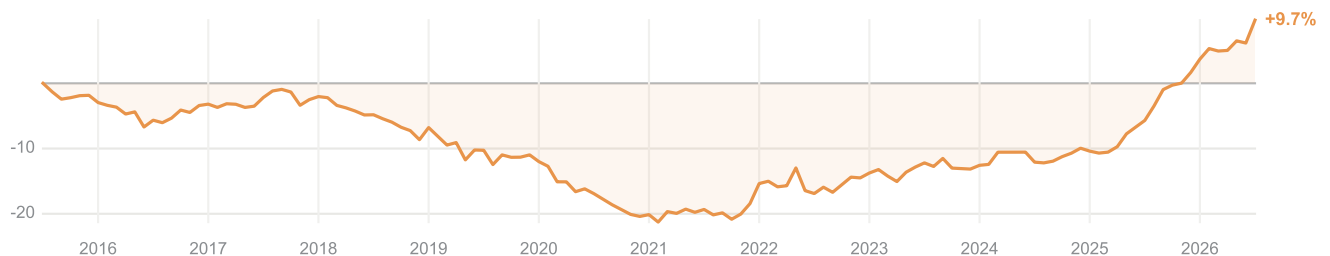
CAGR +7.1%



066 ● 1Y Chg in Cash Flow to Price

Historical Growth

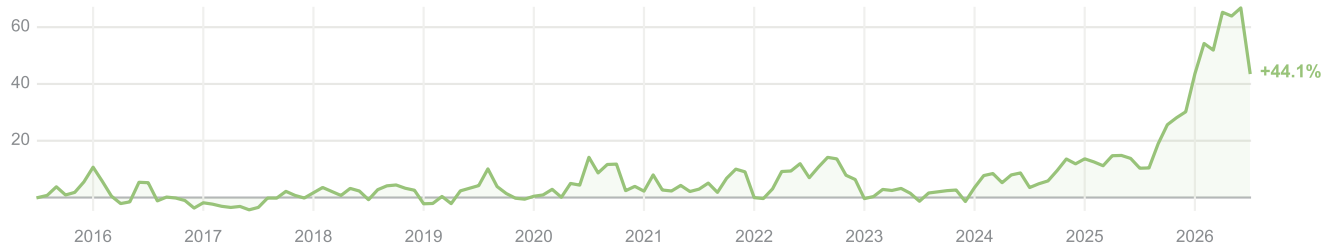
CAGR +7.1%



067 ● 50 Day to 200 Day Stock Price

Price Momentum

CAGR +7.1%



068 ● Adjusted Book to Enterprise Value

Valuation

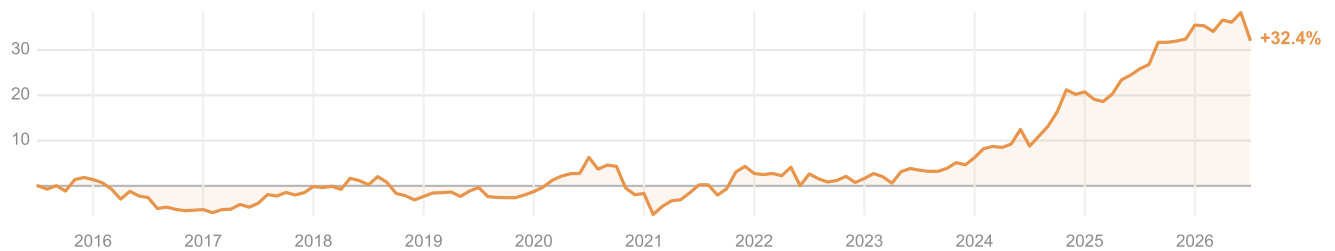
CAGR +7.0%



069 ● 1Y Chg in Asset Adjusted EPS

Historical Growth

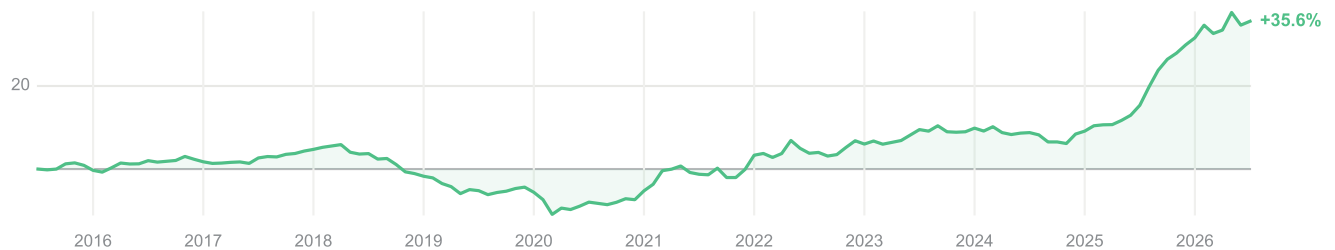
CAGR +7.0%



070 ● Earnings Surprise

Analyst Expectations

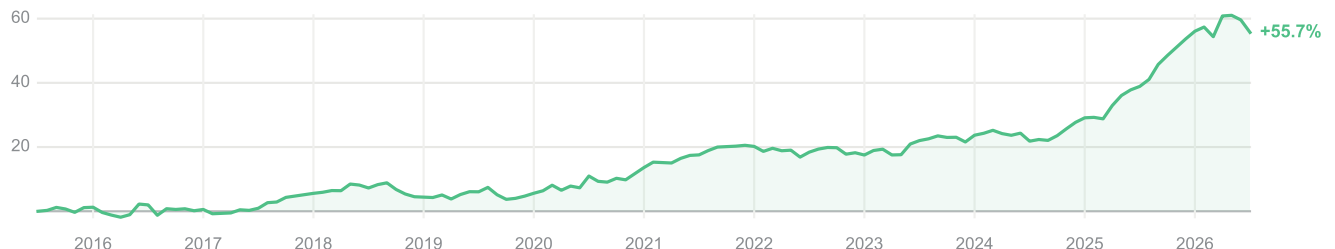
CAGR +7.0%



071 ● Street Revision Magnitude

Analyst Expectations

CAGR +7.0%



072 ● Cash Flow to Invested Capital

Earnings Quality

CAGR +7.0%



073 ● Sales to Invested Capital

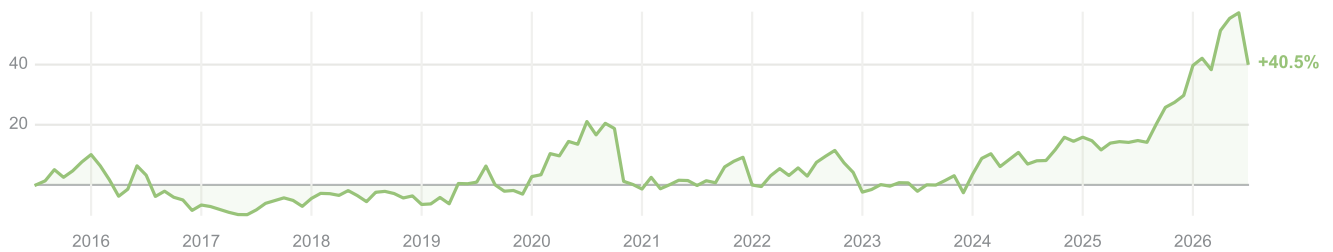
Earnings Quality

CAGR +7.0%

**074 ● 39W Lagged Return**

Price Momentum

CAGR +7.0%

**075 ● Operating Cash Flow to Invested Capital**

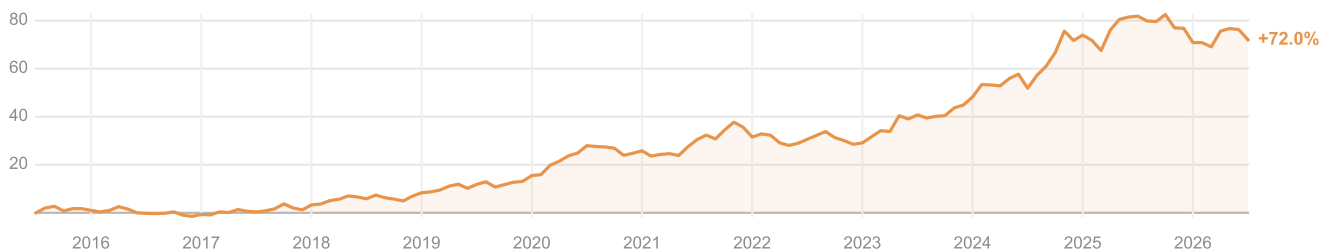
Earnings Quality

CAGR +7.0%

**076 ● Adjusted 3Y Chg in Asset Adjusted Free Cash Flow**

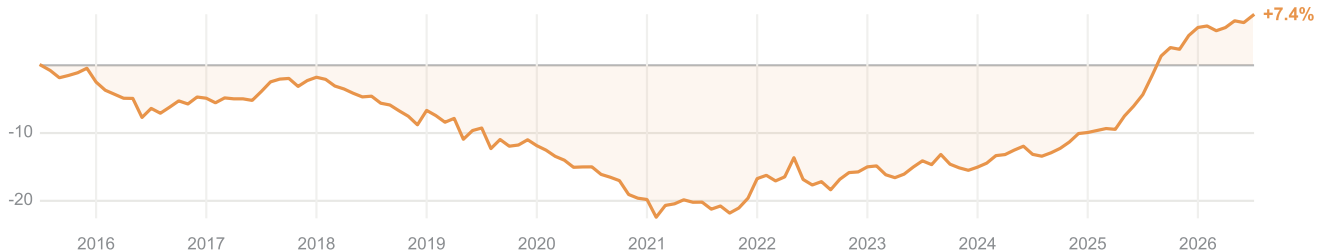
Historical Growth

CAGR +7.0%

**077 ● 1Y Chg in Earnings to Price**

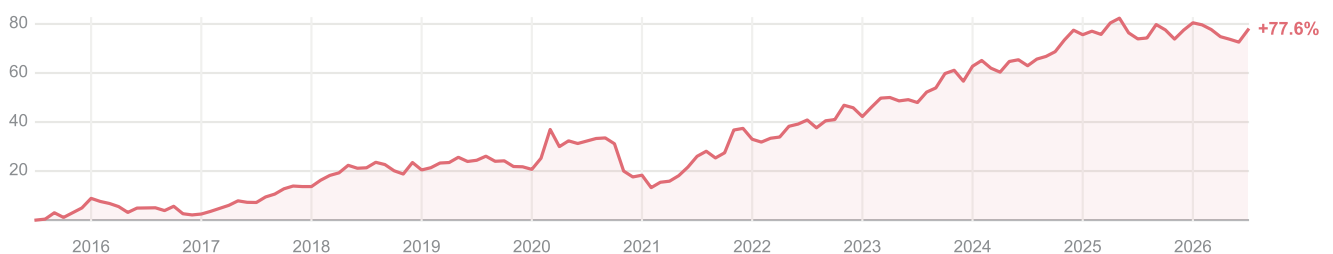
Historical Growth

CAGR +7.0%

**078 ● Operating Cash Flow to Asset**

Earnings Quality

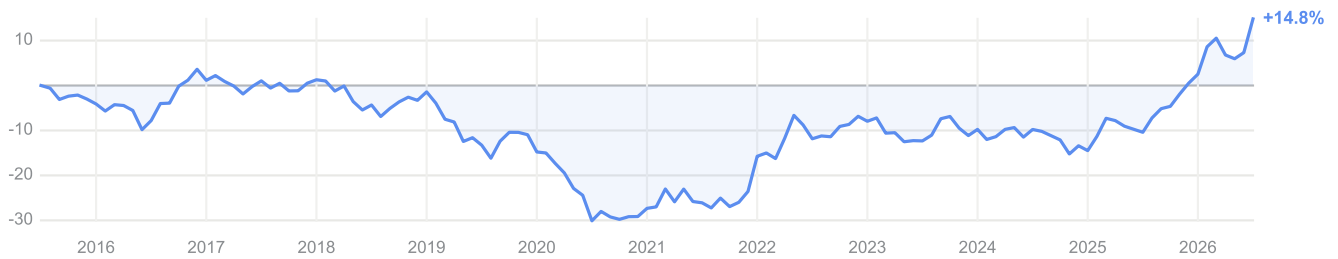
CAGR +6.9%



079 ● **EBITDA to Enterprise Value**

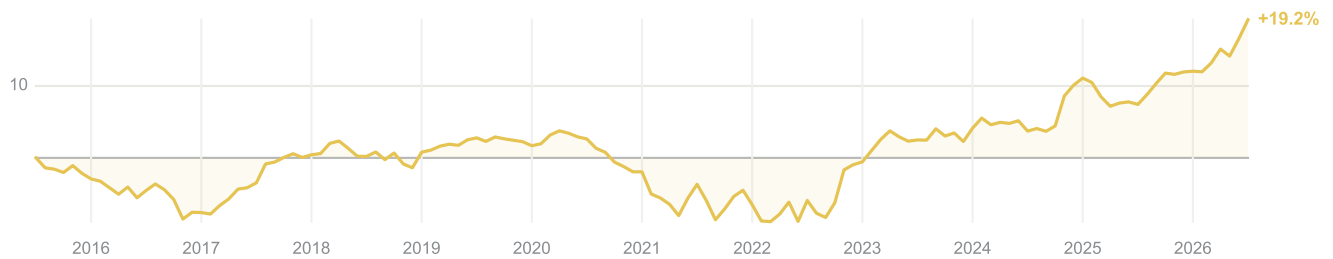
Valuation

CAGR +6.8%

080 ● **1Y Chg in Stock Buybacks**

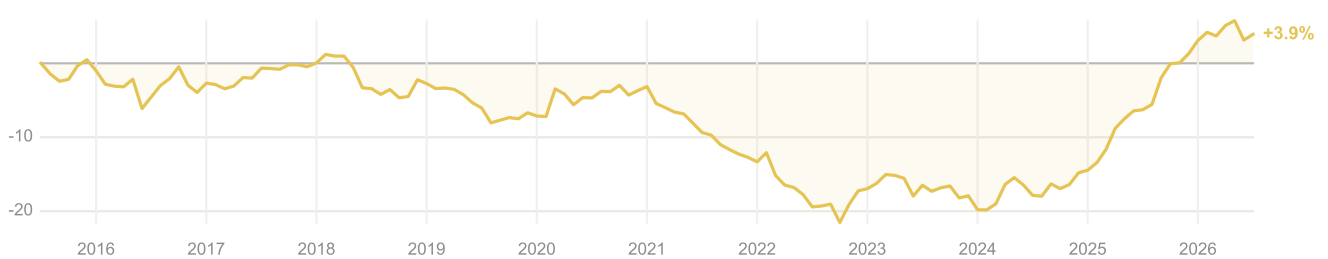
Capital Efficiency

CAGR +6.8%

081 ● **Long Term Debt to Assets Ratio**

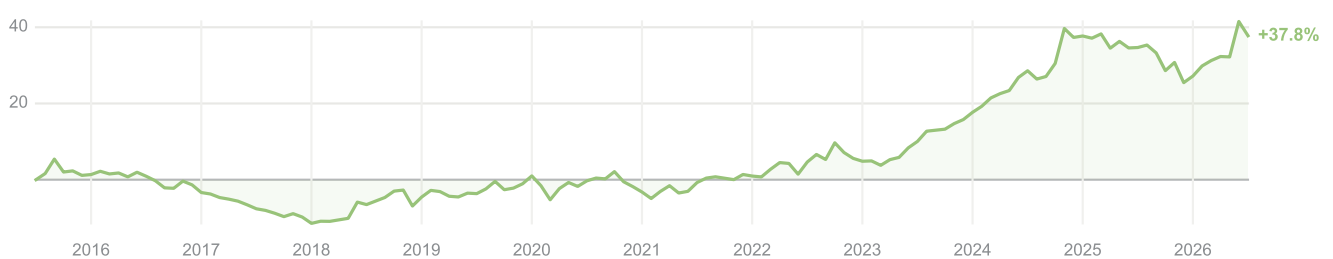
Capital Efficiency

CAGR +6.8%

082 ● **Institution Ownership Level - Active Manager**

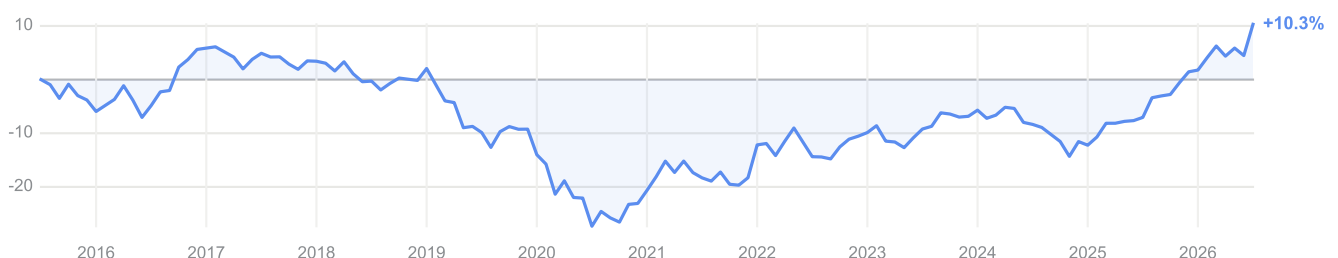
Price Momentum

CAGR +6.8%

083 ● **Operating Cash Flow to Price**

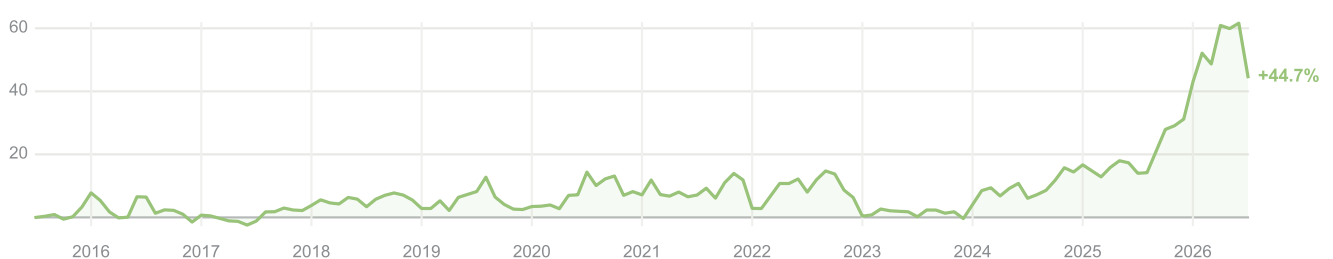
Valuation

CAGR +6.8%

084 ● **15/36 Week Price Ratio**

Price Momentum

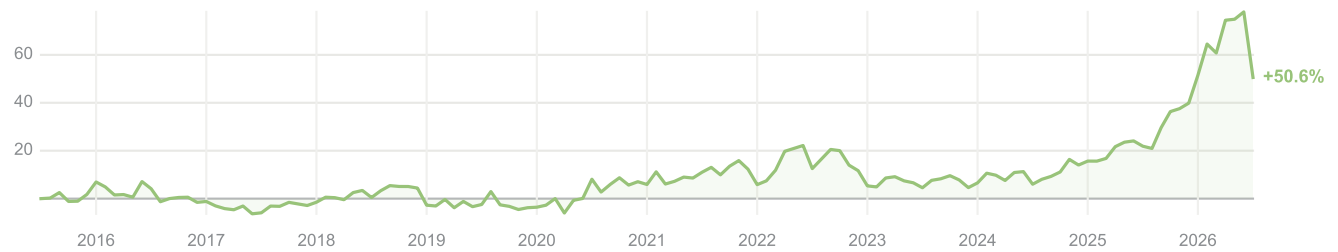
CAGR +6.8%



085 ● **26W Relative Price Strength**

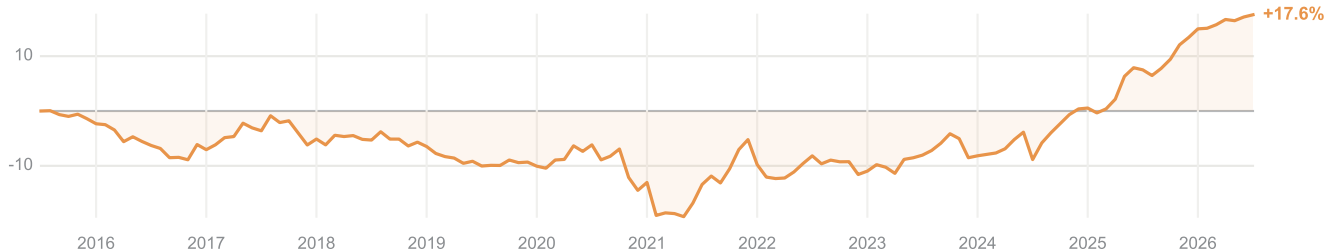
Price Momentum

CAGR +6.8%

086 ● **Adj 1Y Chg in Earnings per Share**

Historical Growth

CAGR +6.7%

087 ● **90 Day Coefficient of Variation**

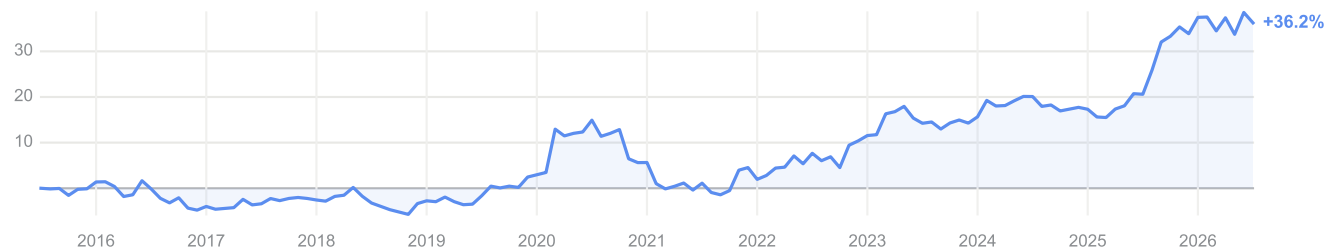
Volatility

CAGR +6.7%

088 ● **Solvency Ratio**

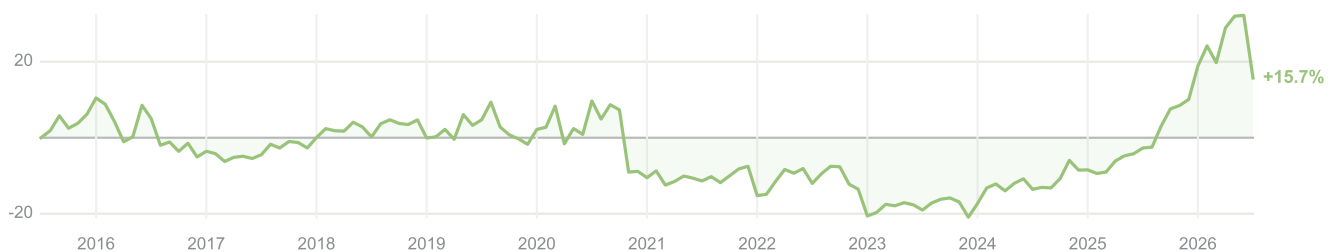
Valuation

CAGR +6.7%

089 ● **9 Month Price Momentum**

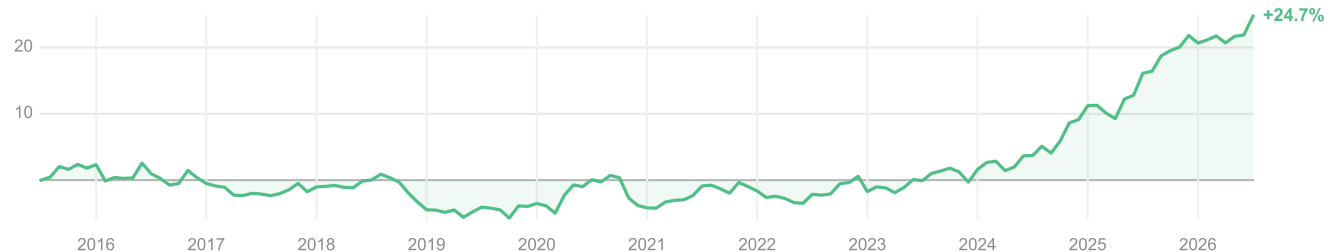
Price Momentum

CAGR +6.6%

090 ● **Standardized Unexpected Earnings**

Analyst Expectations

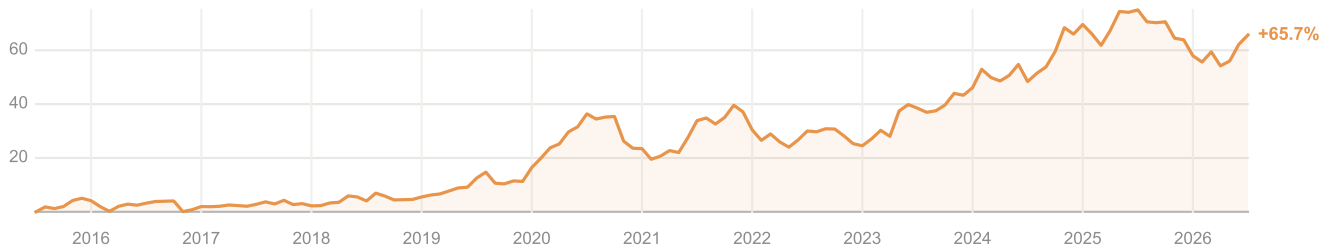
CAGR +6.6%



091 ● 3Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth

CAGR +6.6%



092 ● Adj. EBITDA to Enterprise Value

Valuation

CAGR +6.6%



093 ● Operating Cash Flow to Enterprise Value

Valuation

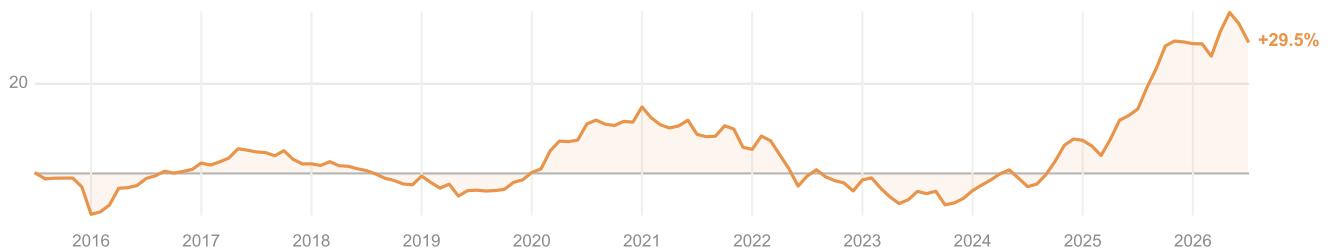
CAGR +6.5%



094 ● 12-Quarter Trendline in Trailing 12-Month Earnings

Historical Growth

CAGR +6.5%



095 ● Cash Ratio

Earnings Quality

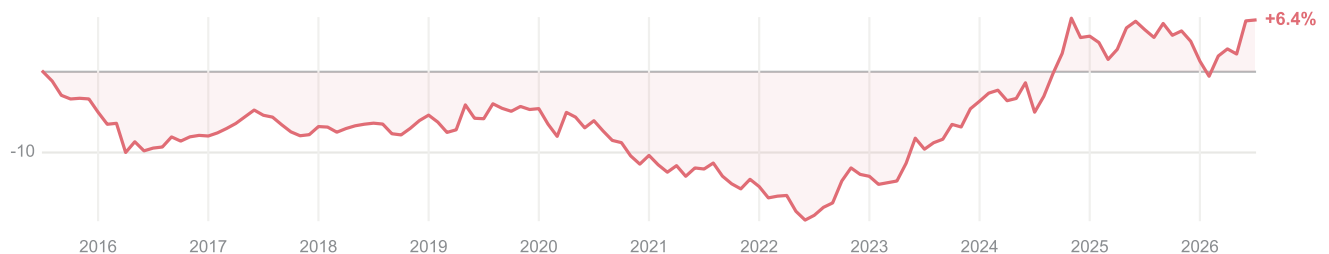
CAGR +6.5%



096 ● Inventory Turnover

Earnings Quality

CAGR +6.4%



097 ● **Book to Price**

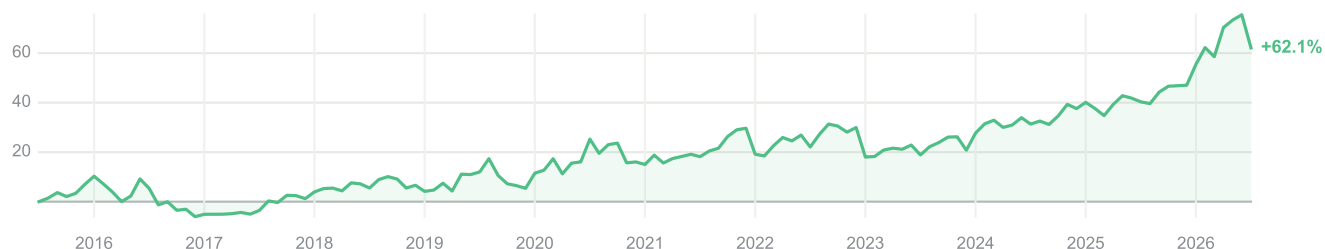
Valuation

CAGR +6.4%

098 ● **6M Chg in Target Price**

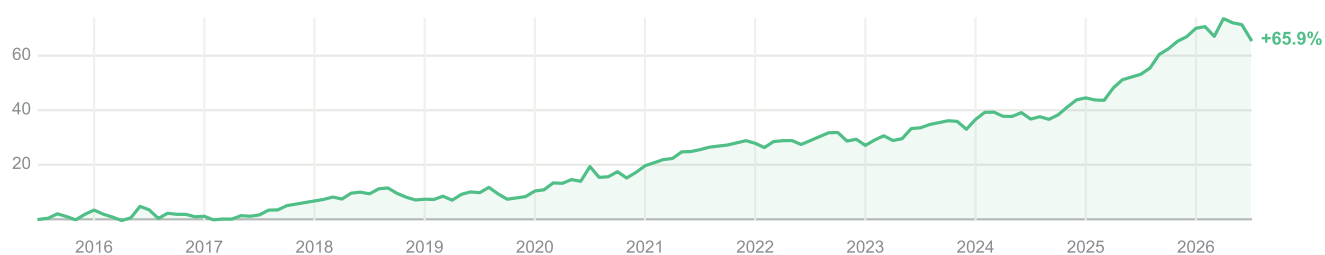
Analyst Expectations

CAGR +6.4%

099 ● **Adjusted Revision Magnitude**

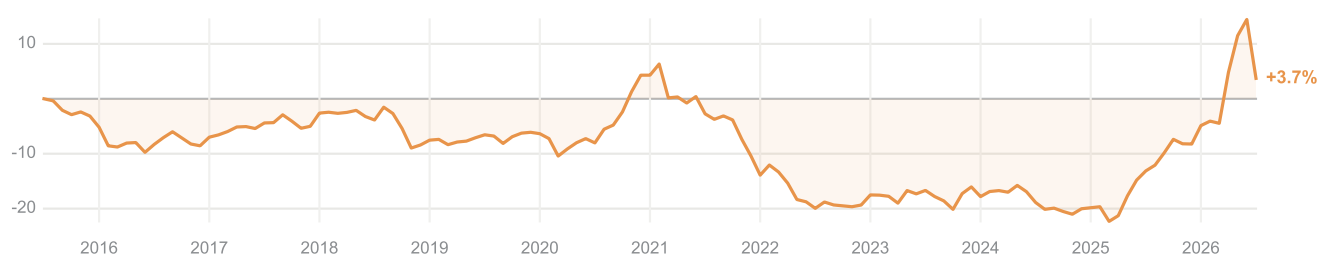
Analyst Expectations

CAGR +6.4%

100 ● **Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**

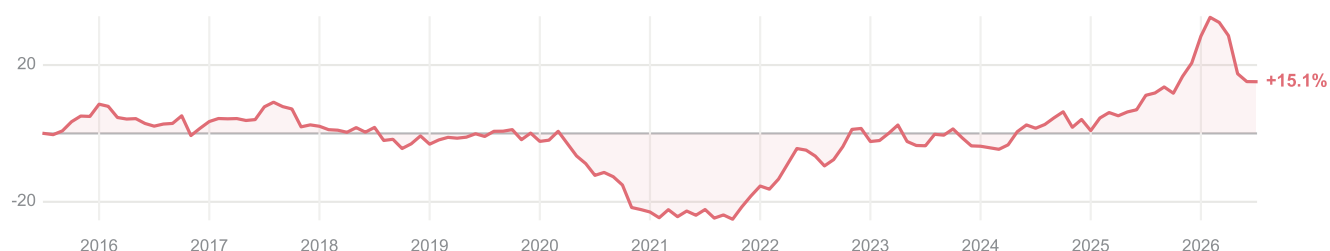
Historical Growth

CAGR +6.4%

101 ● **EBITDA Margin**

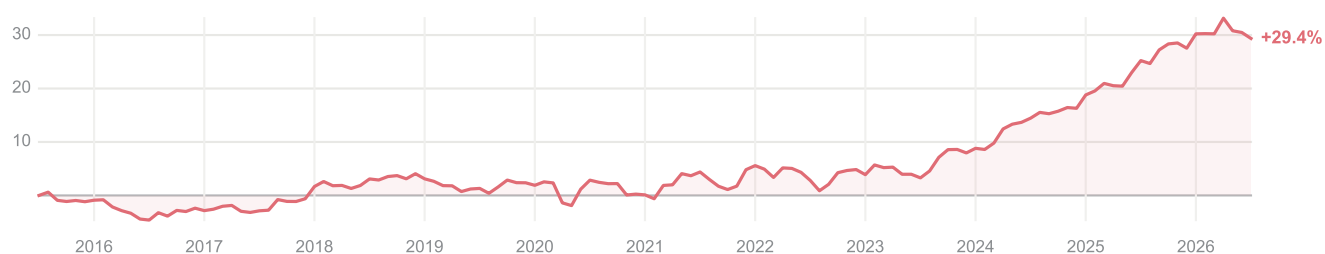
Earnings Quality

CAGR +6.3%

102 ● **Capital Expenditure to Sales**

Earnings Quality

CAGR +6.3%



103 ● Free Cash Flow to Enterprise Value

Valuation

CAGR +6.3%

**104 ● 1M Realized Price Volatility**

Volatility

CAGR +6.3%

**105 ● EBITDA to Price**

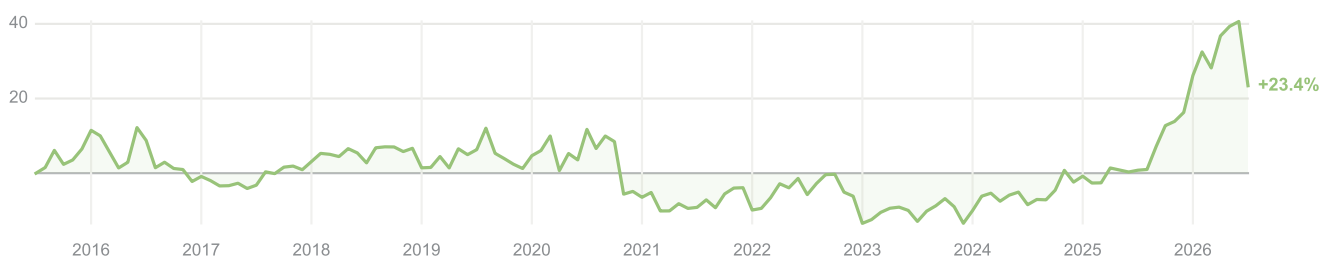
Valuation

CAGR +6.3%

**106 ● 4W to 52W Price Oscillator**

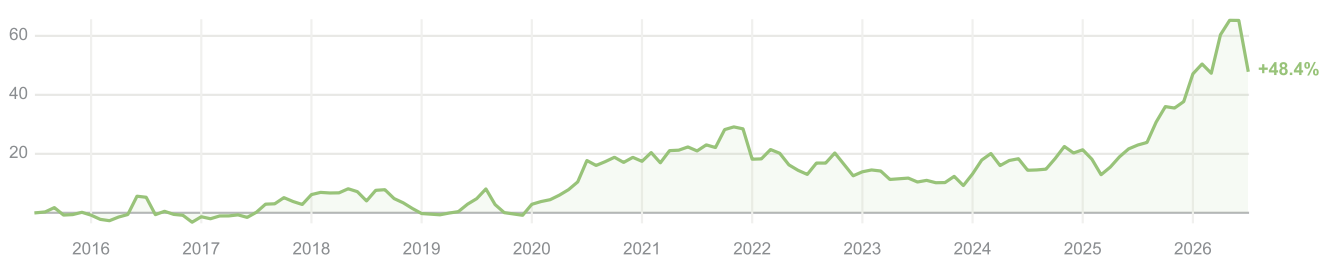
Price Momentum

CAGR +6.3%

**107 ● Percent Above 260-Day Low**

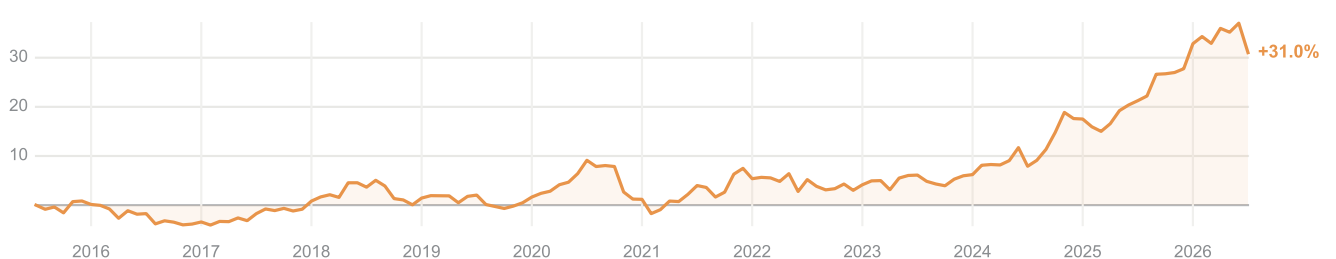
Price Momentum

CAGR +6.3%

**108 ● 1Y Chg in Asset Adjusted Cash Flow**

Historical Growth

CAGR +6.2%

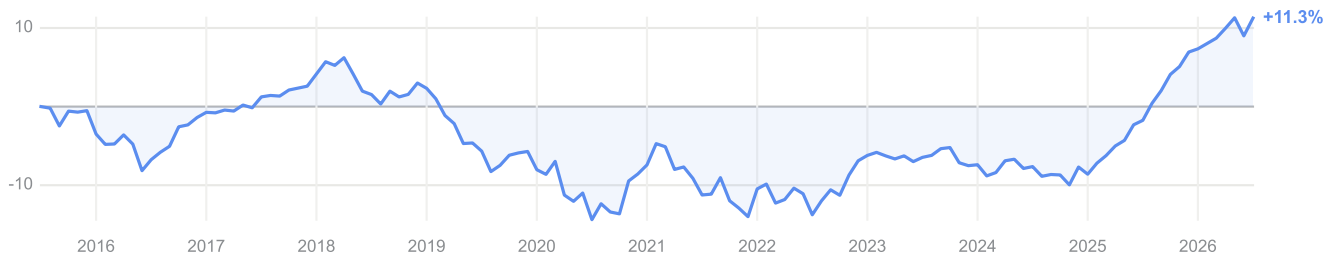


109 ● **Growth Flow to Price**

Valuation CAGR +6.2%

110 ● **Cash to EV Ratio**

Valuation CAGR +6.2%

111 ● **Year over Year Change in SGA to Sales**

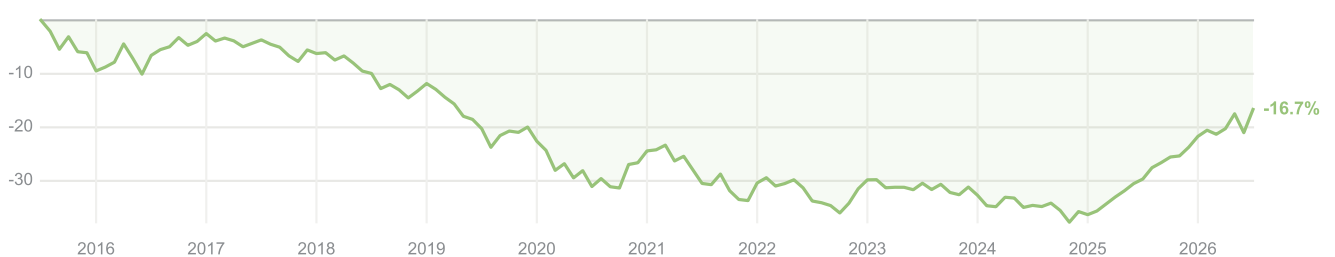
Earnings Quality CAGR +6.2%

112 ● **Adjusted 6 Month Average Share Turn Over**

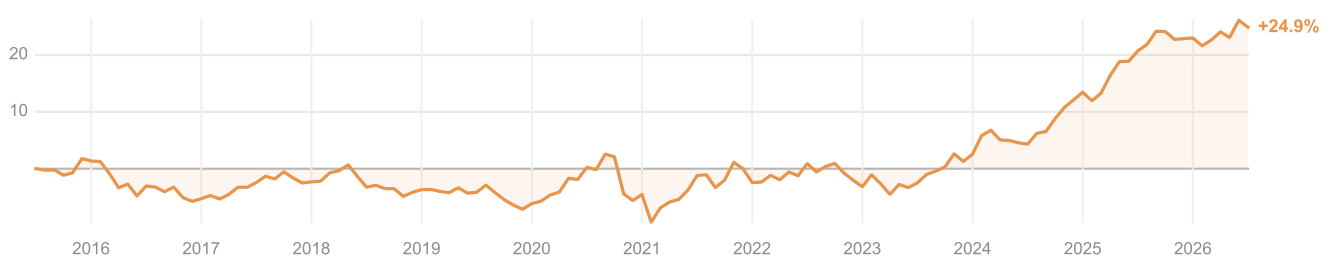
Price Momentum CAGR +6.2%

113 ● **Log of Unadjusted Stock Price**

Price Momentum CAGR +6.1%

114 ● **1Y Chg in Operating Cash Flow per Share**

Historical Growth CAGR +6.1%



115 ● **Analyst Dispersion for FY1 EPS**

Analyst Expectations

CAGR +6.1%

116 ● **Cash Flow Return on Invested Capital**

Capital Efficiency

CAGR +6.1%

117 ● **1Y Chg in Asset Adjusted Free Cash Flow**

Historical Growth

CAGR +6.0%

118 ● **Stability of Change in Number of Holders**

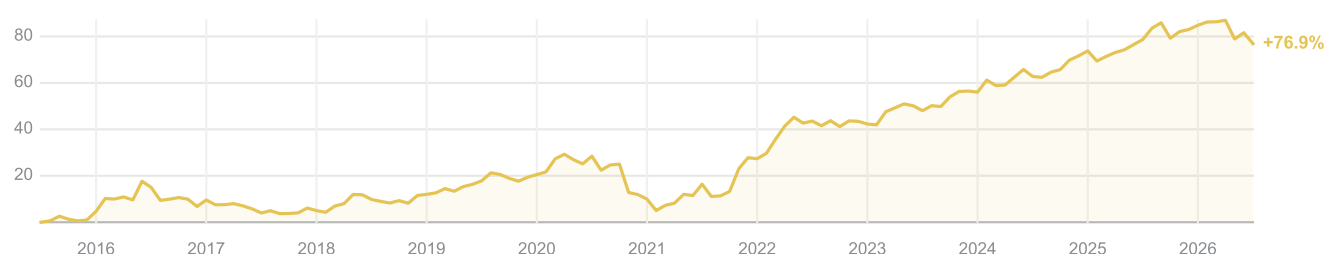
Price Momentum

CAGR +6.0%

119 ● **Operating Cash Flow Ratio**

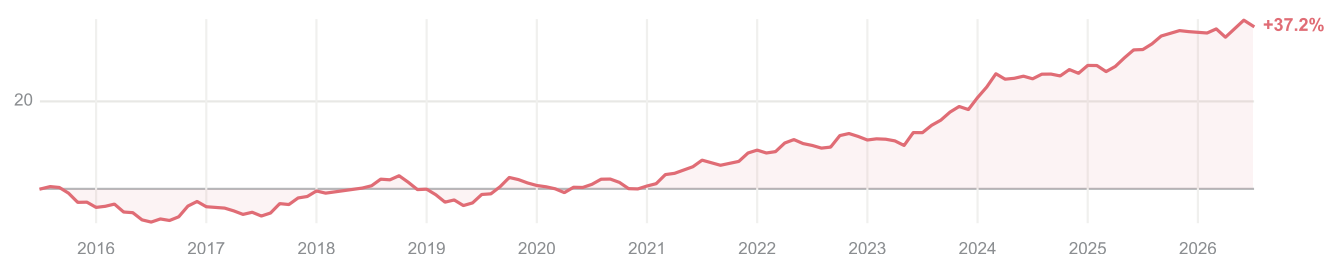
Capital Efficiency

CAGR +6.0%

120 ● **Equity Turnover**

Earnings Quality

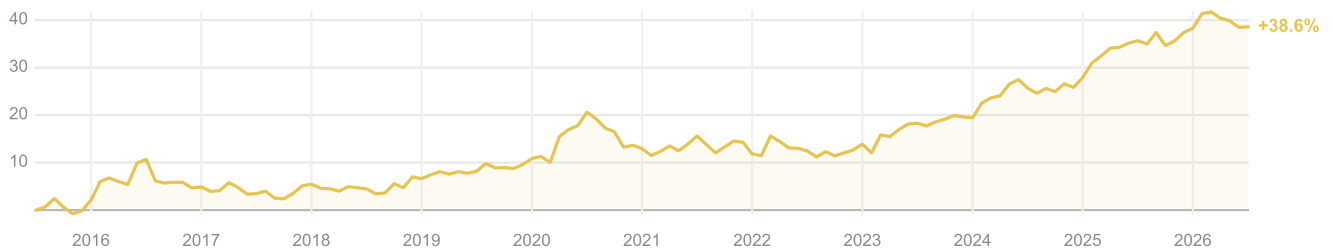
CAGR +6.0%



121 ● Receivables Turnover Ratio

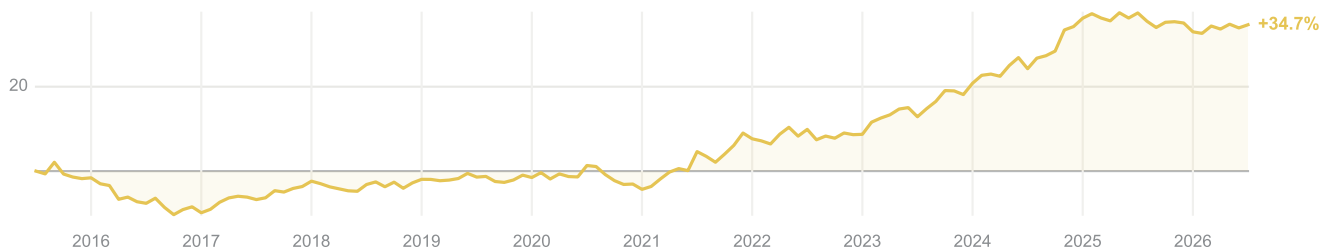
Capital Efficiency

CAGR +6.0%

**122 ● Adjusted Interest Coverage Ratio**

Capital Efficiency

CAGR +5.9%

**123 ● 3Y Chg in Free Cash Flow to Price**

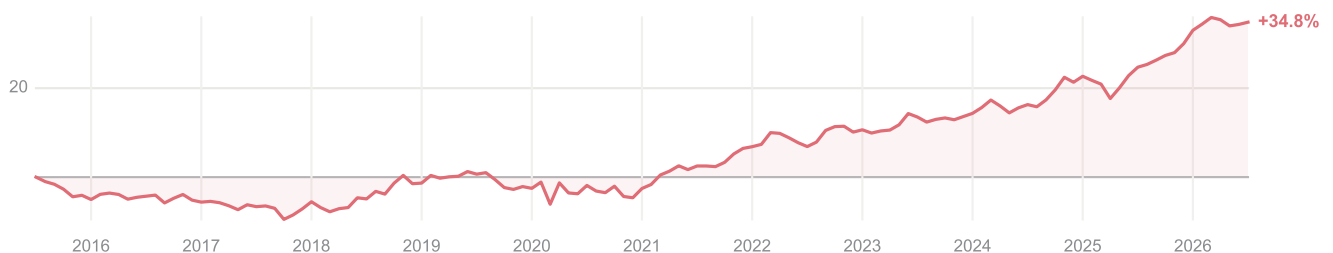
Historical Growth

CAGR +5.9%

**124 ● Working Capital Accruals**

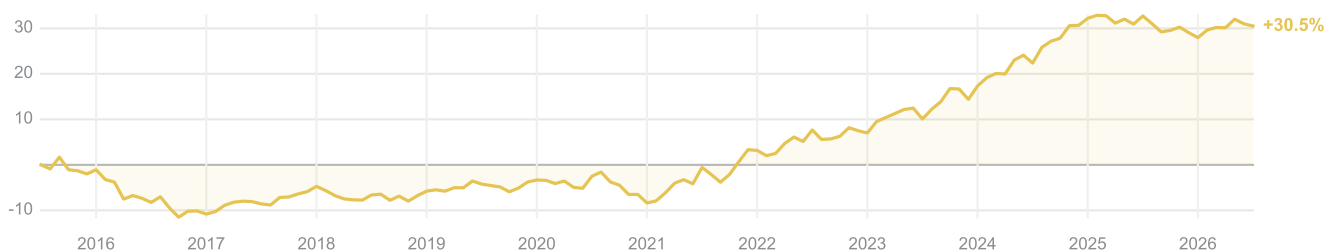
Earnings Quality

CAGR +5.8%

**125 ● Interest Coverage Ratio**

Capital Efficiency

CAGR +5.8%

**126 ● Forward Free Cash Flow to Price Ratio**

Valuation

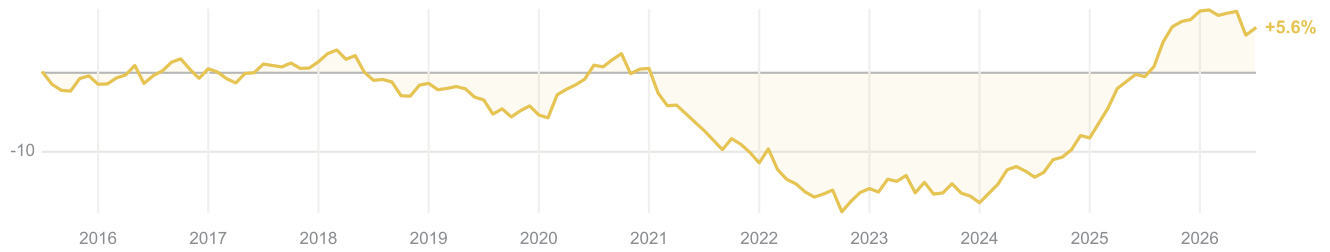
CAGR +5.8%



127 ● **Debt to Assets Ratio**

Capital Efficiency

CAGR +5.8%

128 ● **3Y Chg in Asset Adjusted Cash Flow**

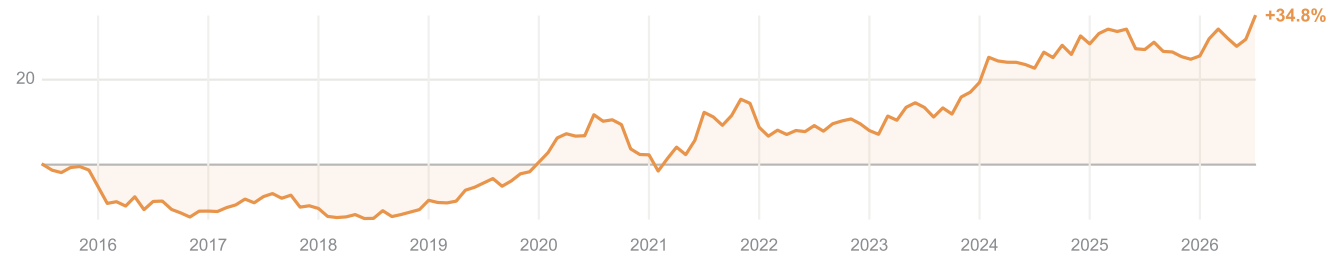
Historical Growth

CAGR +5.8%

129 ● **3Y Chg in Cash Flow per Share**

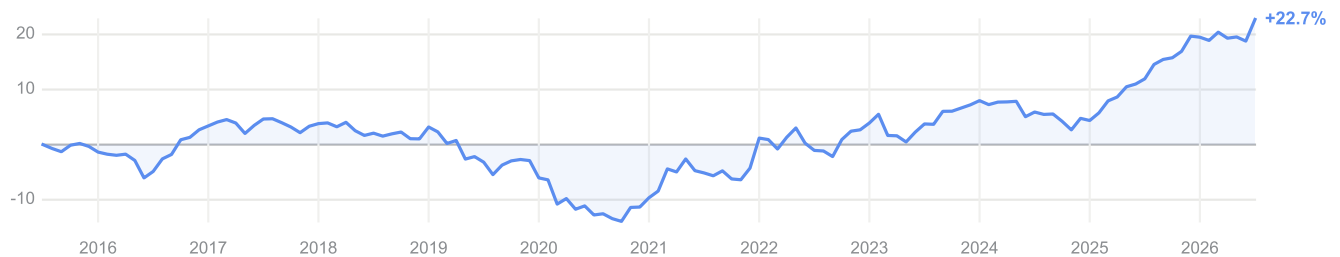
Historical Growth

CAGR +5.8%

130 ● **Free Cash Flow to Price**

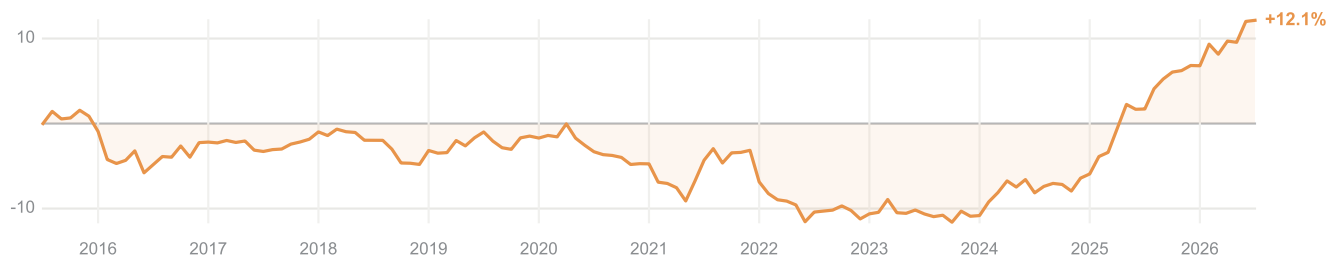
Valuation

CAGR +5.7%

131 ● **Dividend per Share Growth**

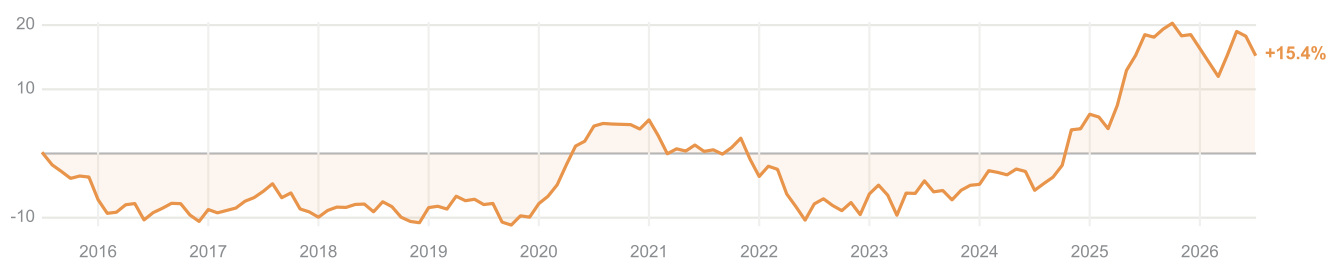
Historical Growth

CAGR +5.7%

132 ● **12-Quarter Trendline in Trailing 12-Month Cash Flows**

Historical Growth

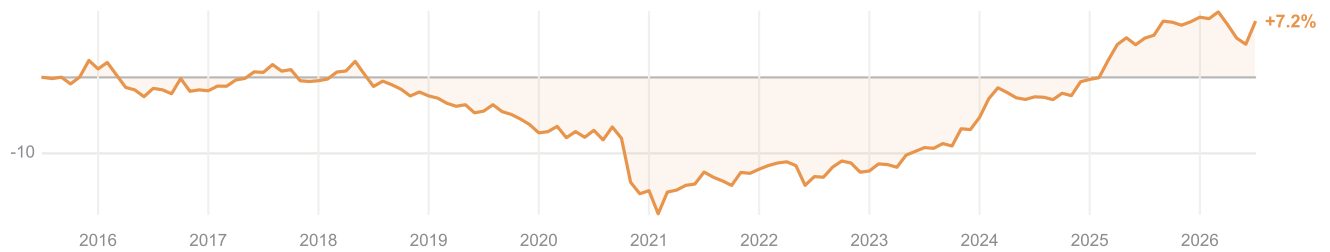
CAGR +5.6%



133 ● 1Y Chg in Operating Cash Flow to Price

Historical Growth

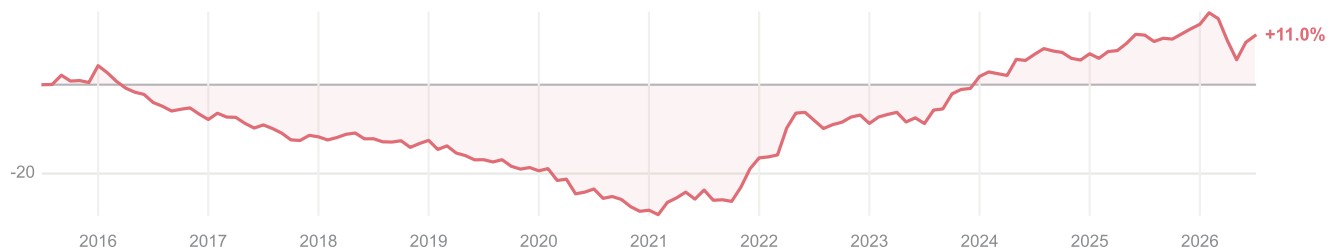
CAGR +5.5%



134 ● Cash Burn Rate

Earnings Quality

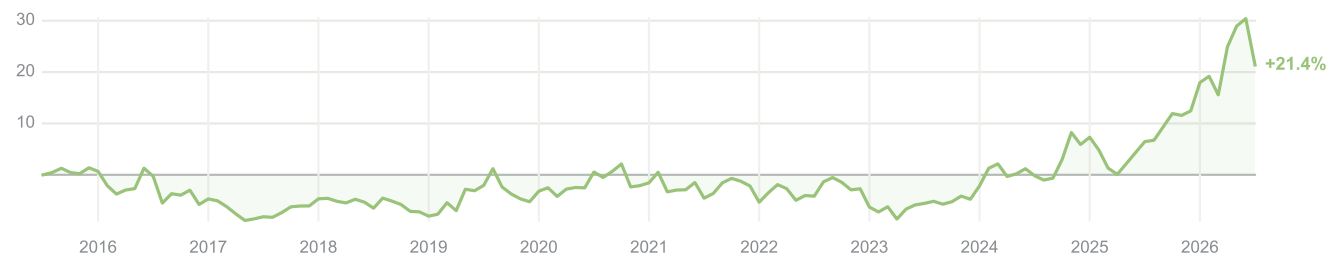
CAGR +5.5%



135 ● 52-Week Volume Price Trend with 20-Day Lag

Price Momentum

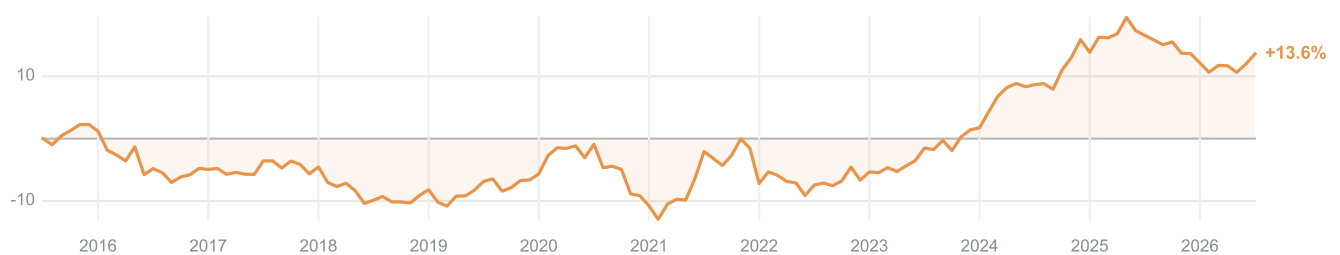
CAGR +5.5%



136 ● 3Y Average Annual Earnings Growth

Historical Growth

CAGR +5.5%



137 ● Depreciation to Capex Ratio

Earnings Quality

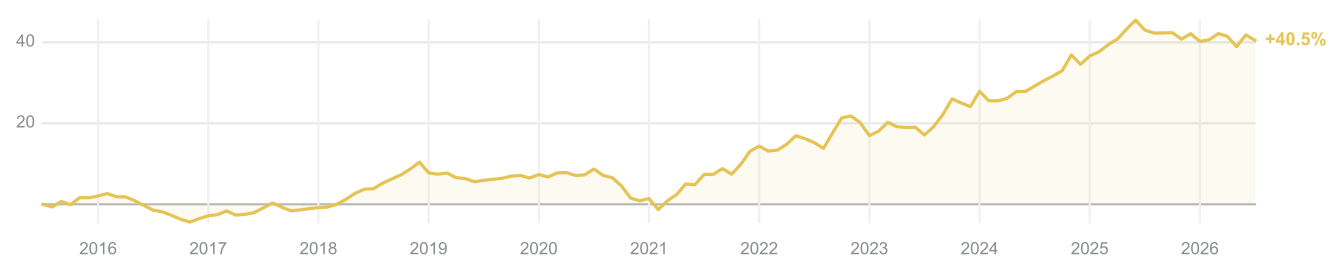
CAGR +5.5%

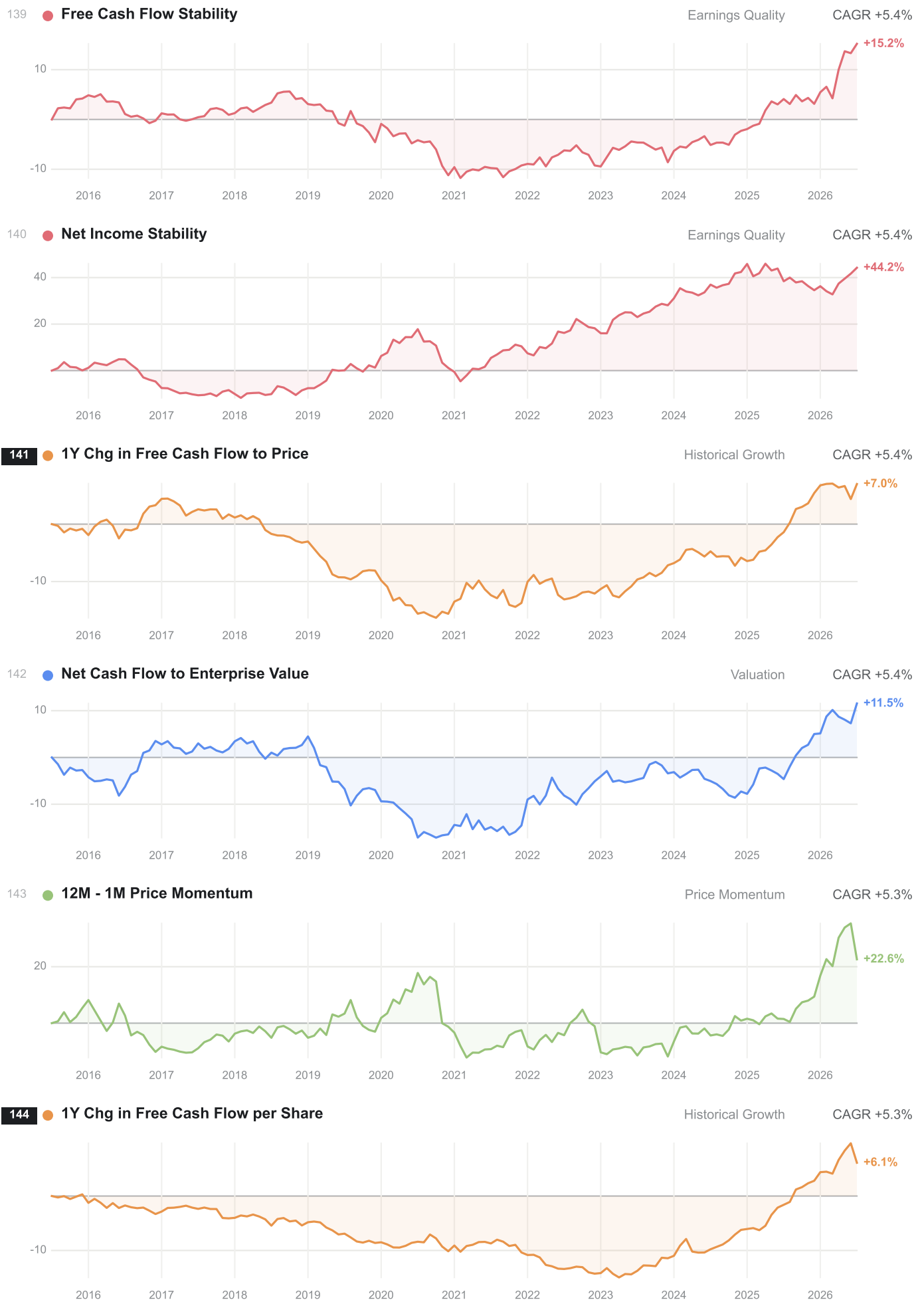


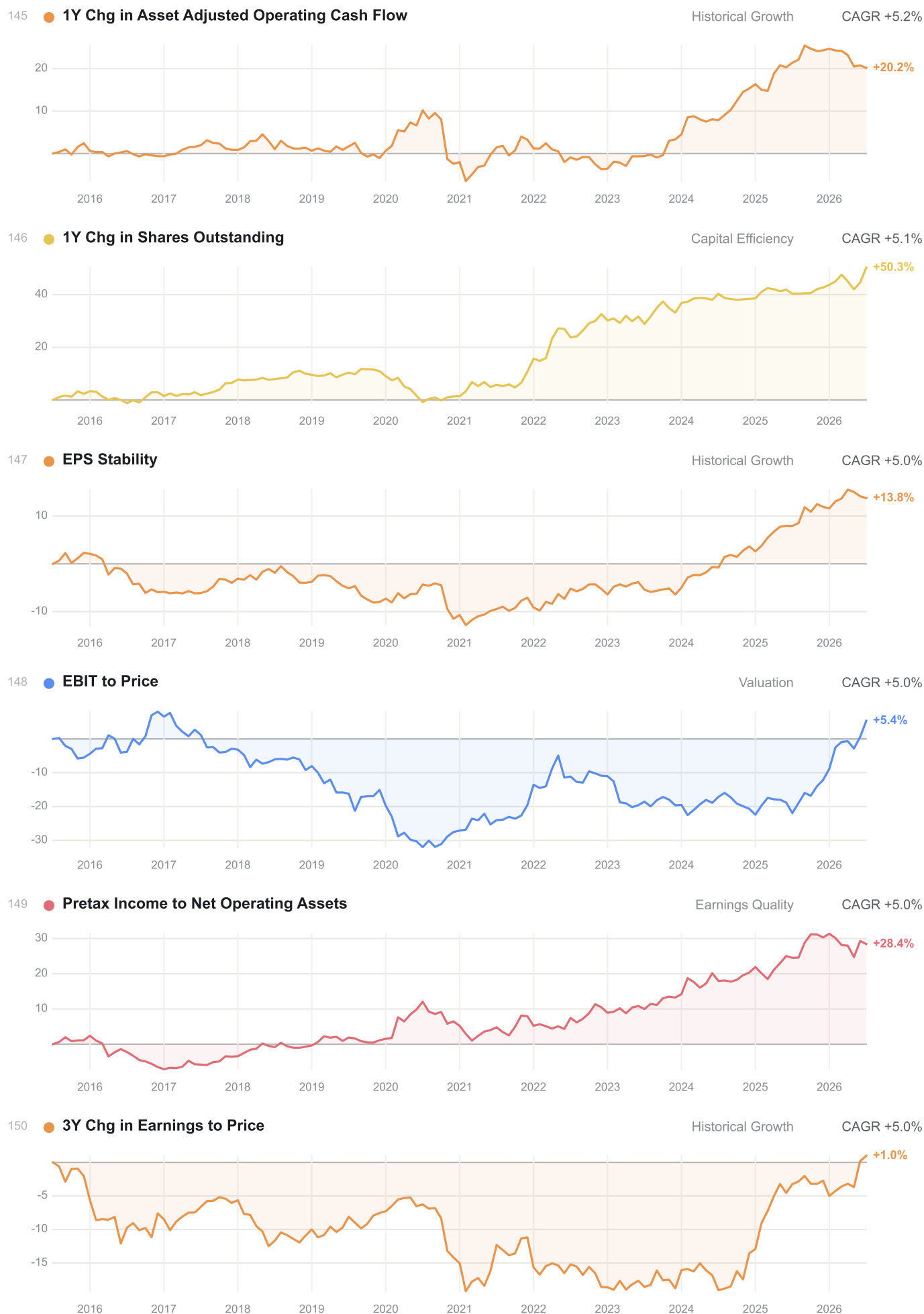
138 ● Capital Acquisition Ratio

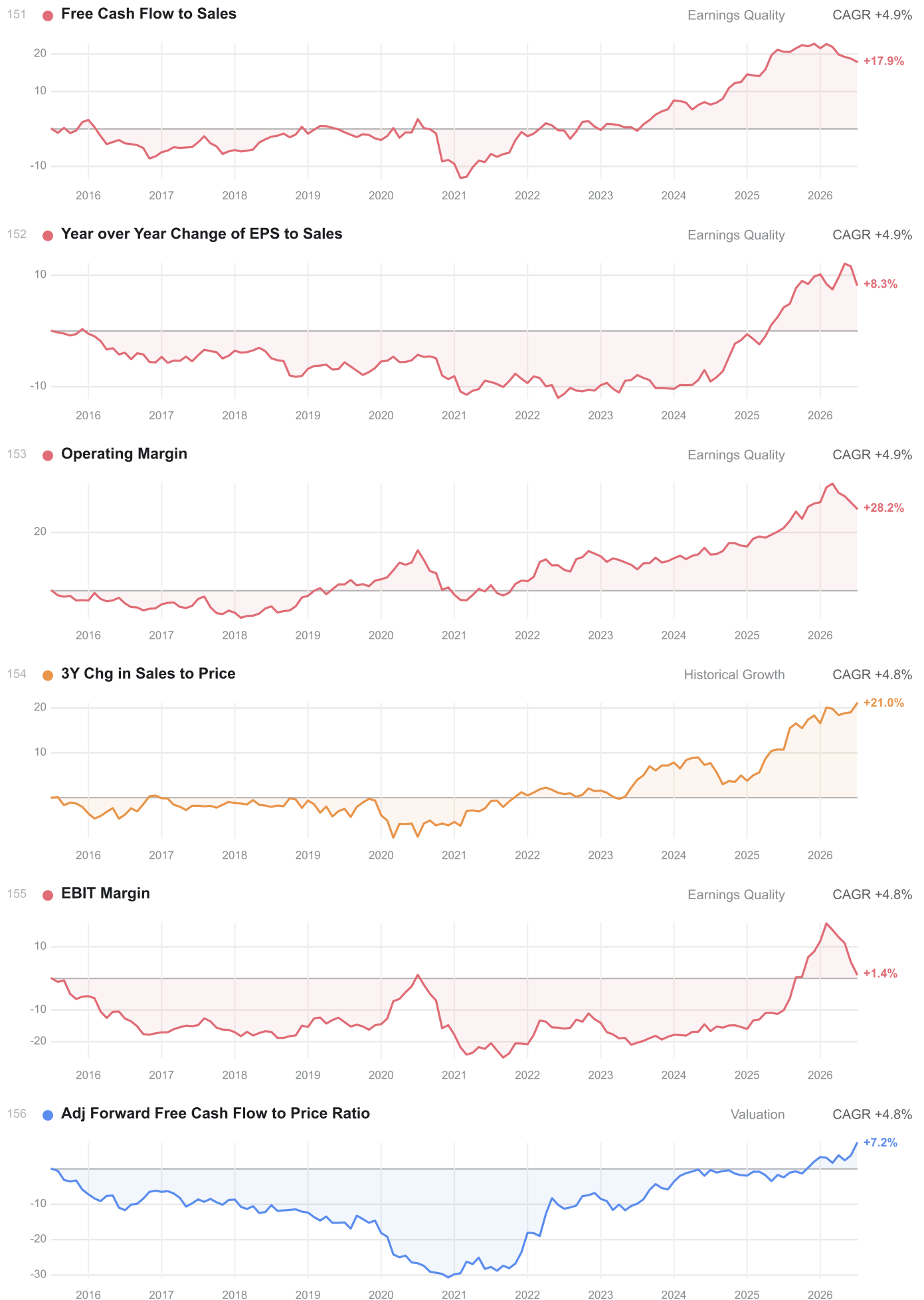
Capital Efficiency

CAGR +5.5%





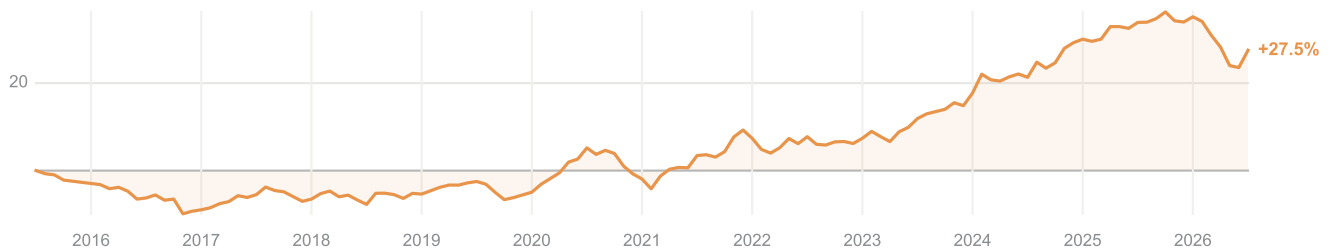




157 ● **3Y Chg in Free Cash Flow per Share**

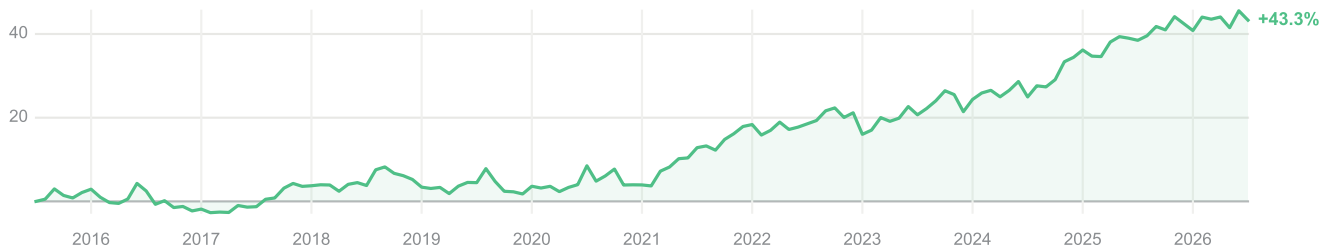
Historical Growth

CAGR +4.8%

158 ● **Number of EPS FY1 Revisions**

Analyst Expectations

CAGR +4.7%

159 ● **Working Capital to Assets Ratio**

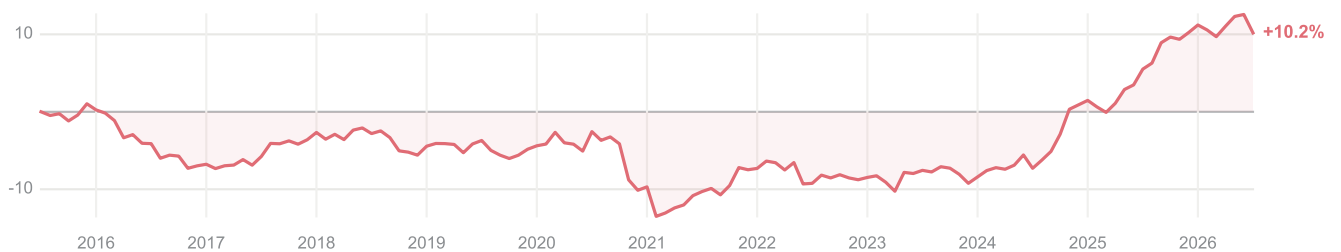
Earnings Quality

CAGR +4.7%

160 ● **Change EPS to Sales**

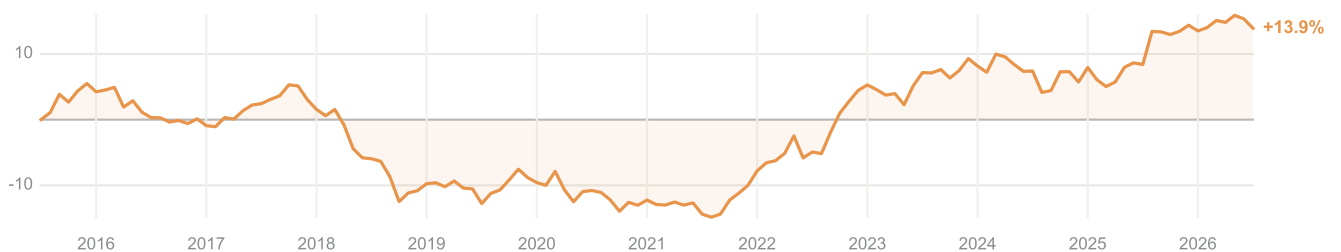
Earnings Quality

CAGR +4.7%

161 ● **Num of Quarters of Declines in Receivables and Inventories/TTM Sales**

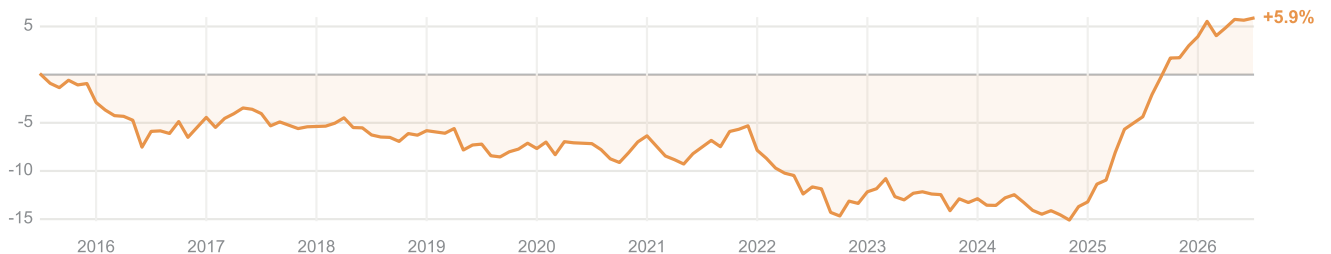
Historical Growth

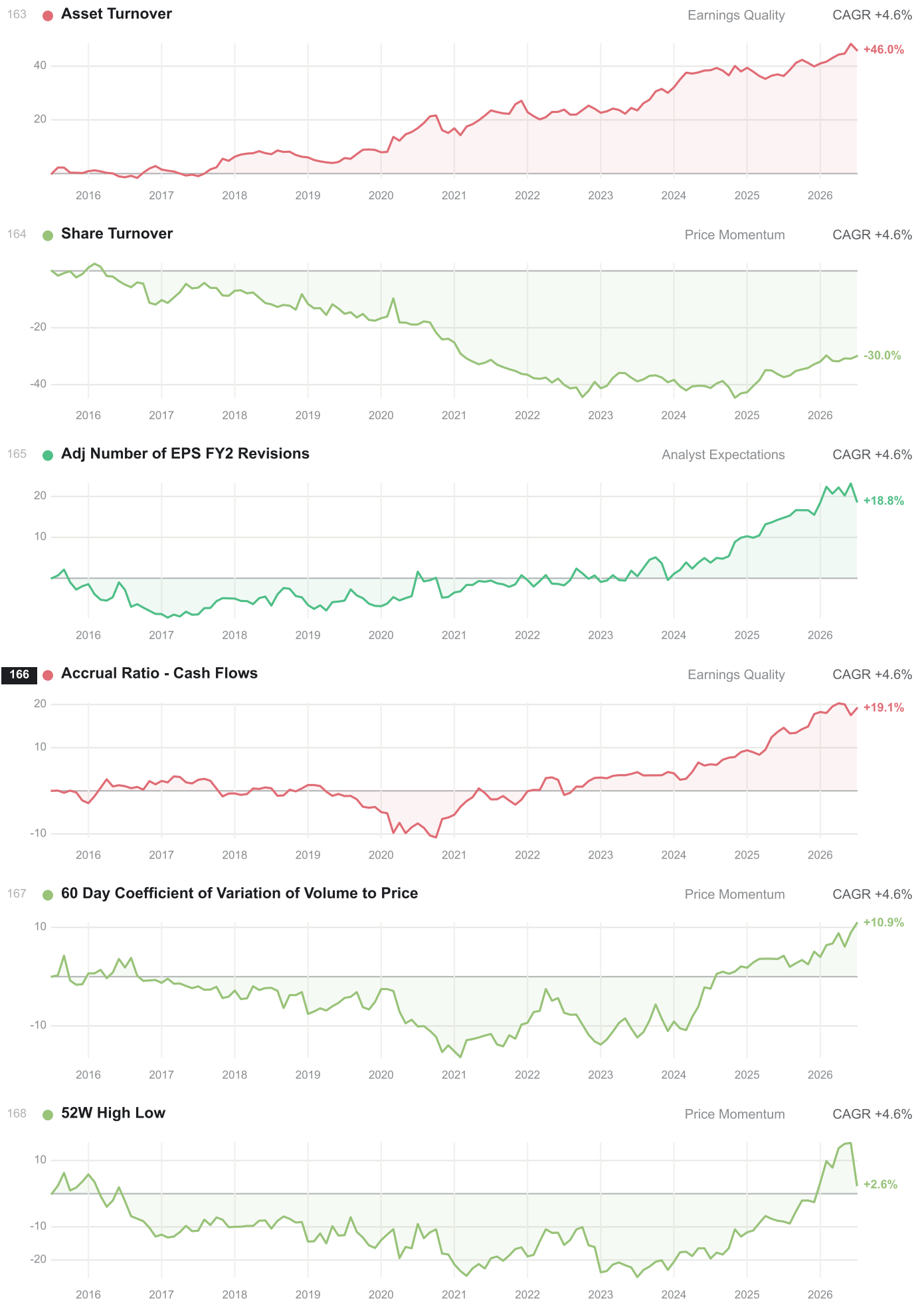
CAGR +4.7%

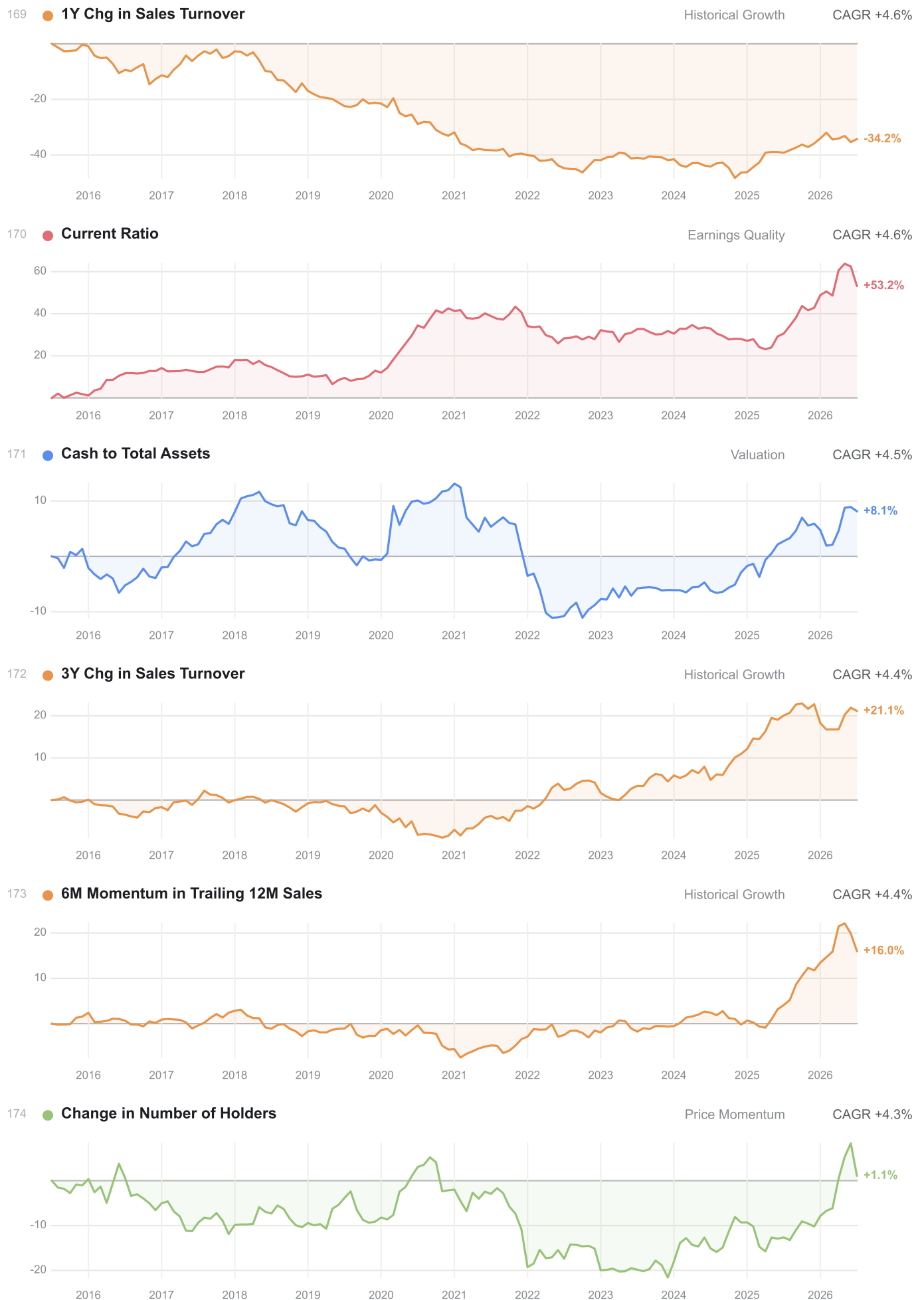
162 ● **Dividend Growth**

Historical Growth

CAGR +4.6%



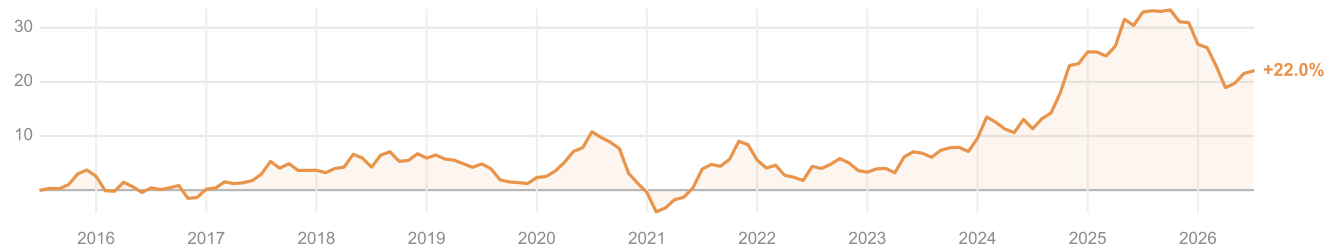




175 ● **3Y Chg in Asset Adjusted Free Cash Flow**

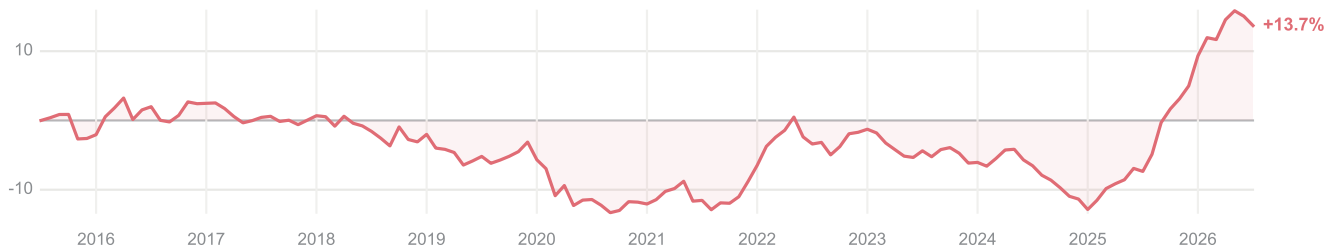
Historical Growth

CAGR +4.3%

176 ● **SG&A to Sales**

Earnings Quality

CAGR +4.2%

177 ● **Foreign Institution Ownership**

Price Momentum

CAGR +4.2%

178 ● **Net Current Assets to Price Ratio**

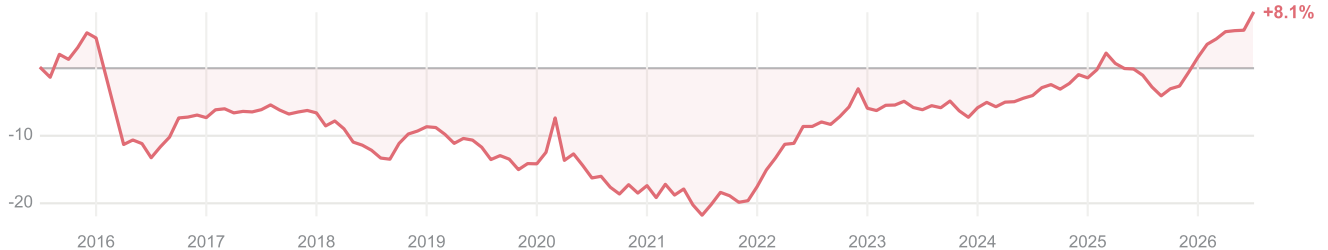
Valuation

CAGR +4.2%

179 ● **Sales to Gross Profit Margin Growth**

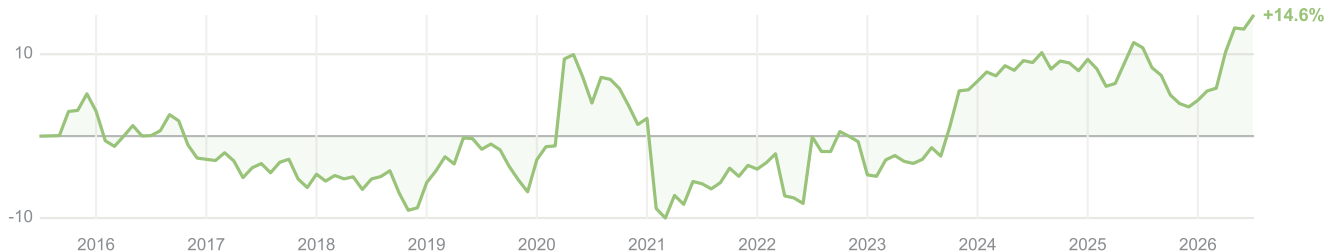
Earnings Quality

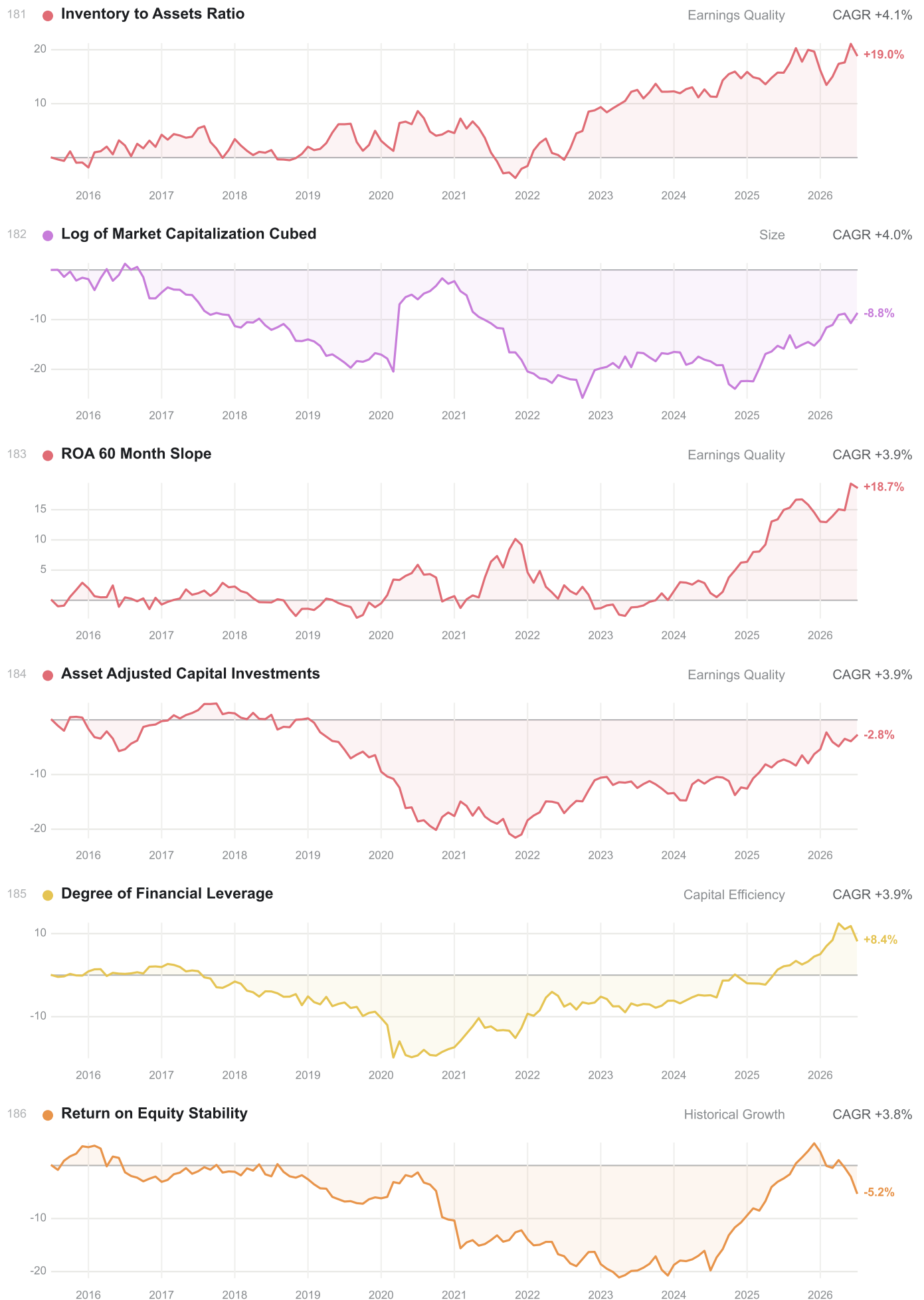
CAGR +4.1%

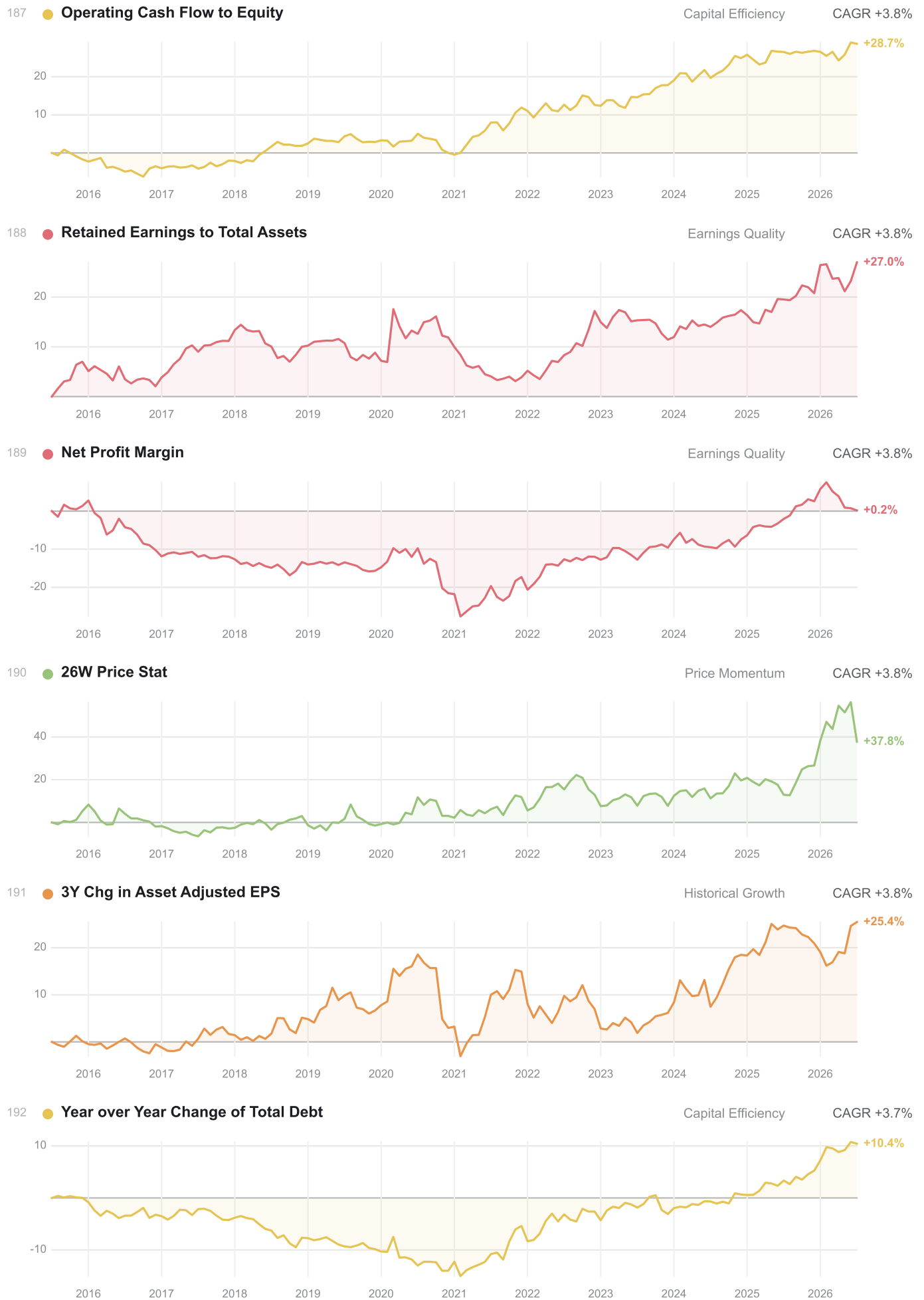
180 ● **Risk Adjusted Relative Strength**

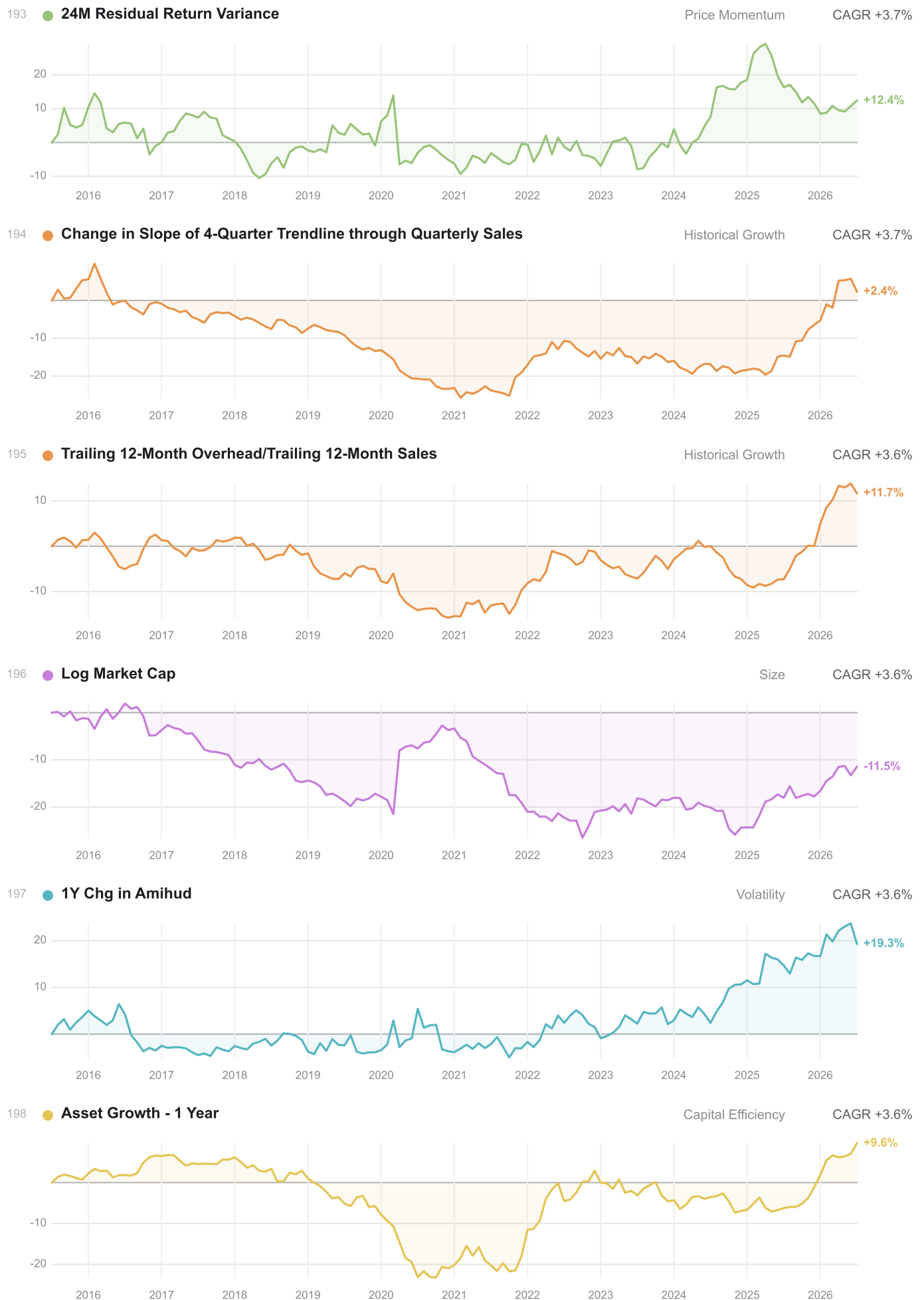
Price Momentum

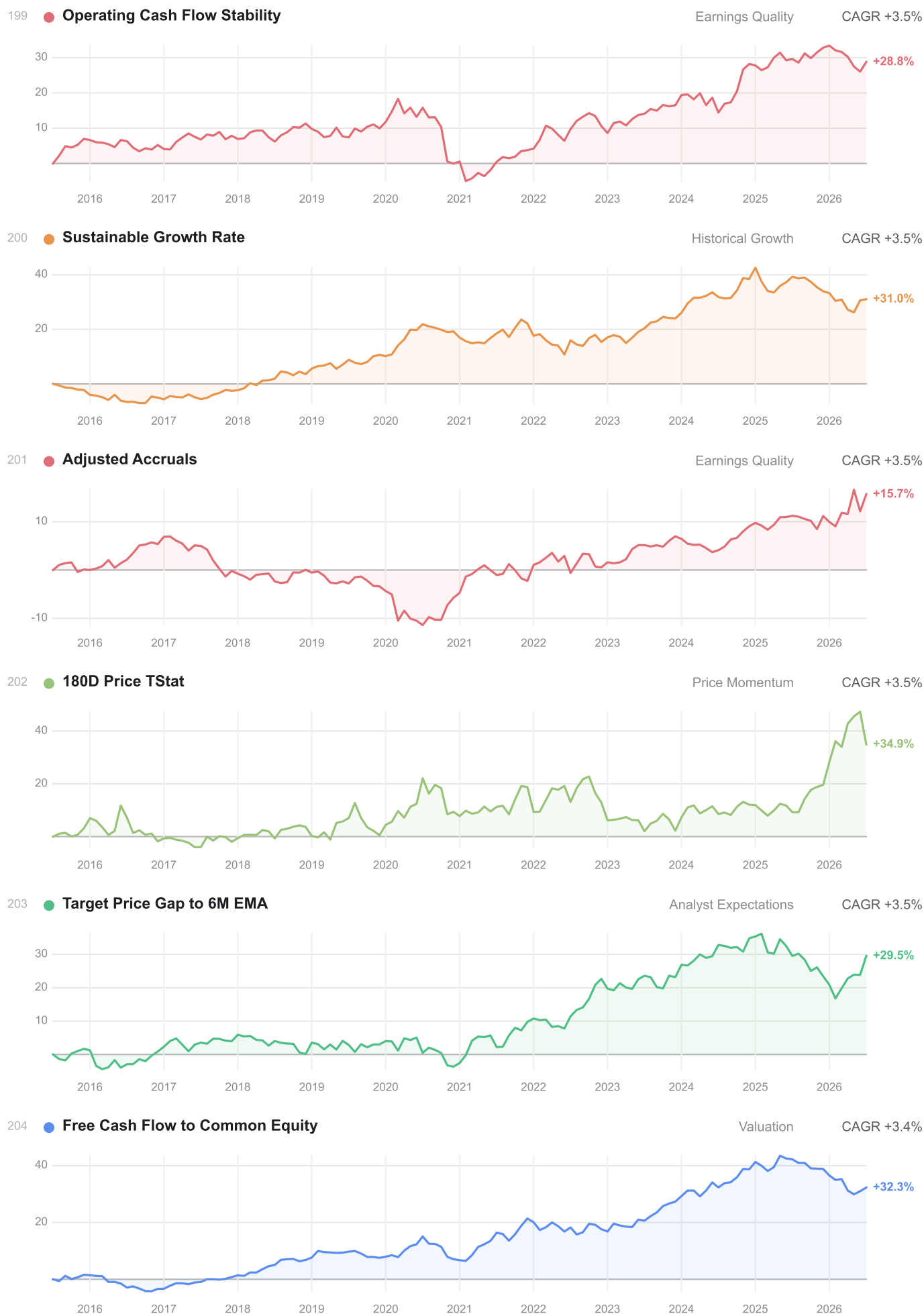
CAGR +4.1%











205 ● 1Y Chg in Earnings per Share

Historical Growth

CAGR +3.4%



206 ● Working Capital to Total Assets

Earnings Quality

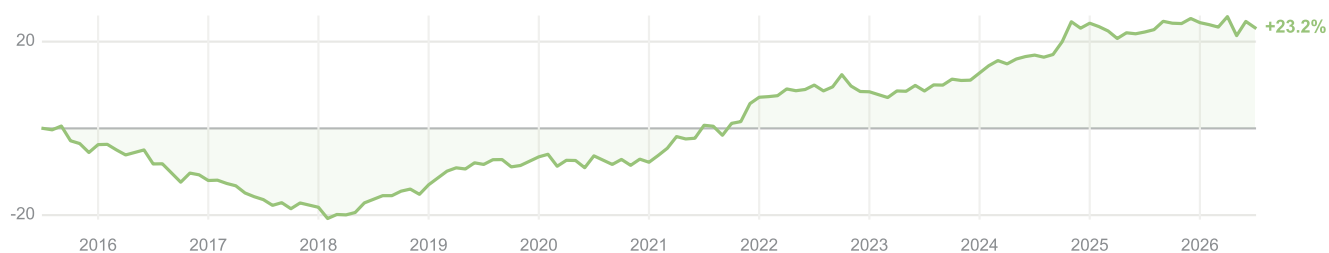
CAGR +3.4%



207 ● Institution Ownership Concentration

Price Momentum

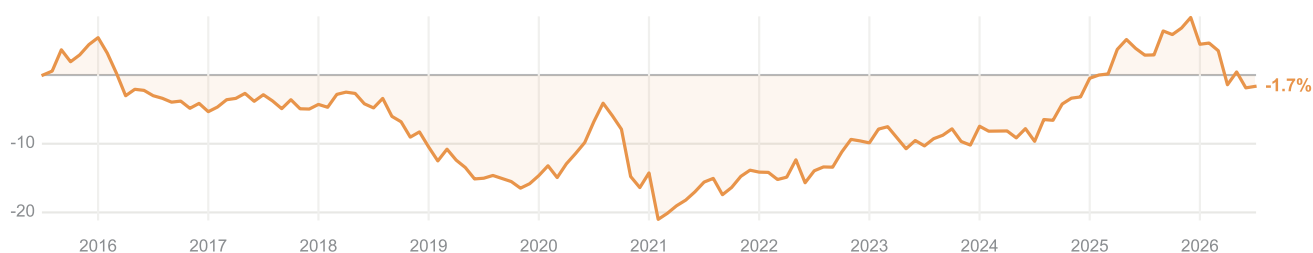
CAGR +3.4%



208 ● Return on Assets Stability

Historical Growth

CAGR +3.3%



209 ● 3Y Chg in Earnings per Share

Historical Growth

CAGR +3.3%



210 ● Accrual Ratio - Balance Sheet

Earnings Quality

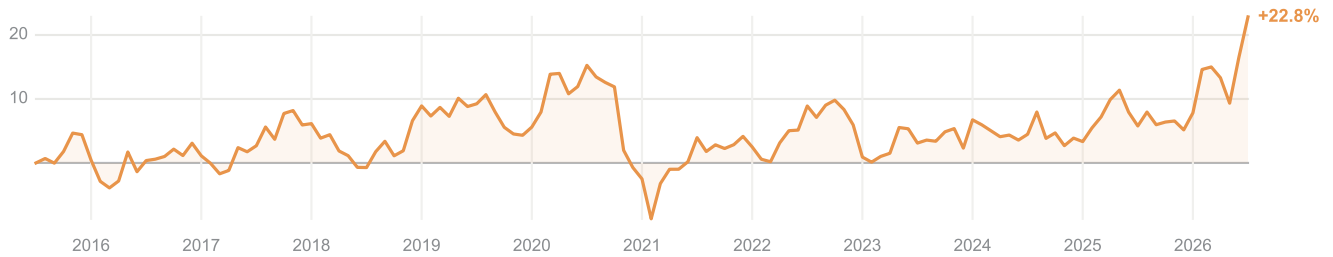
CAGR +3.3%



211 ● 3Y Chg in Operating Margin

Historical Growth

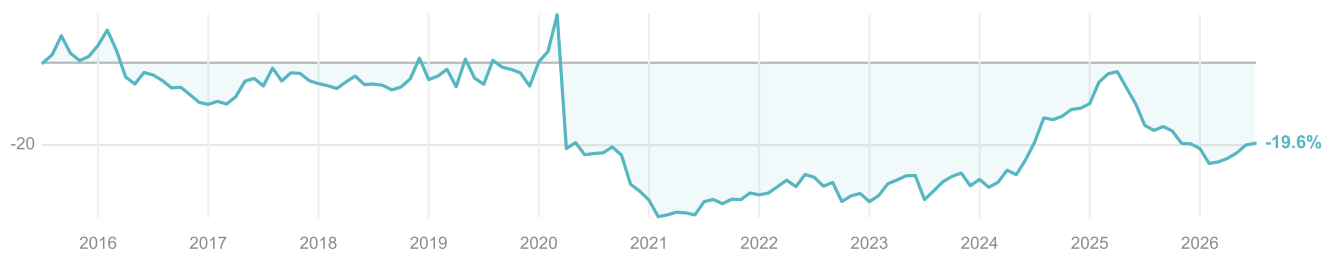
CAGR +3.3%



212 ● 130 Day Minimum Return

Volatility

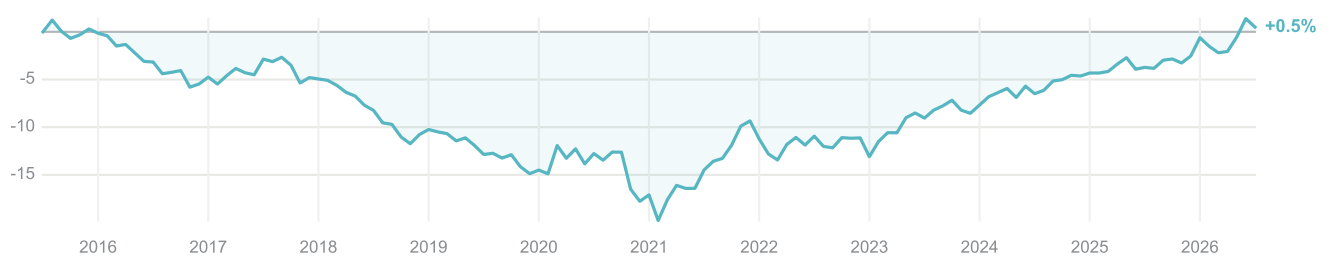
CAGR +3.3%



213 ● 1Y Chg in Debt to Assets

Volatility

CAGR +3.3%



214 ● Ownership Breadth Stability of Institution Ownership by Buyers

Price Momentum

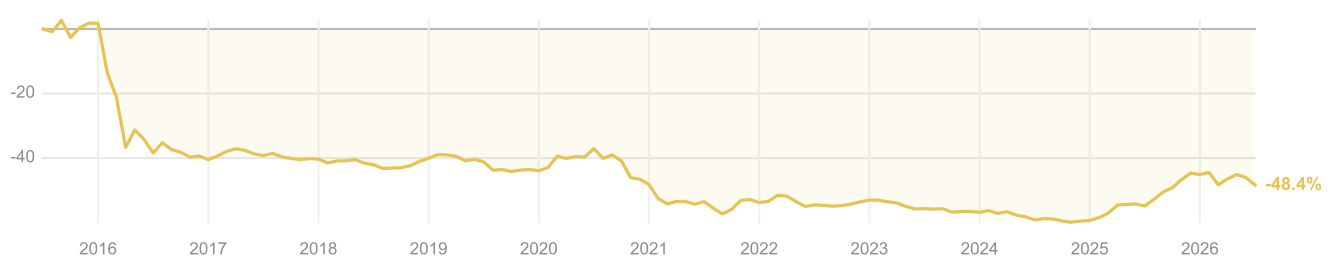
CAGR +3.3%



215 ● Pretax Profit Margin

Capital Efficiency

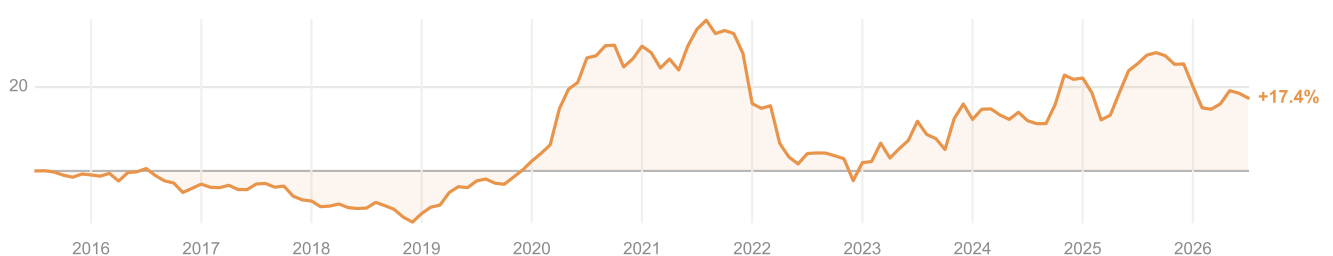
CAGR +3.2%



216 ● 3Y Average Annual Sales Growth

Historical Growth

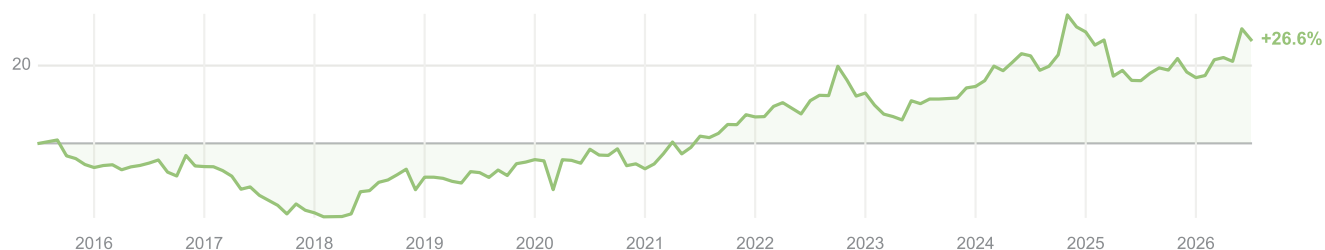
CAGR +3.2%



217 ● **Institution Ownership Level**

Price Momentum

CAGR +3.2%

218 ● **Research & Development Intensity**

Earnings Quality

CAGR +3.2%

219 ● **1Y Chg in Sales to Earnings**

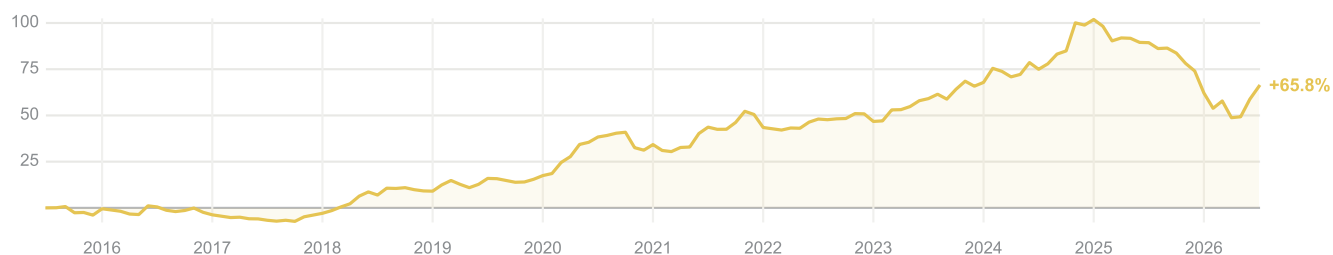
Earnings Quality

CAGR +3.1%

220 ● **Gross Profit to Assets**

Capital Efficiency

CAGR +3.1%

221 ● **Cash Conversion Cycle**

Earnings Quality

CAGR +3.1%

222 ● **Change in Number of Buyers**

Price Momentum

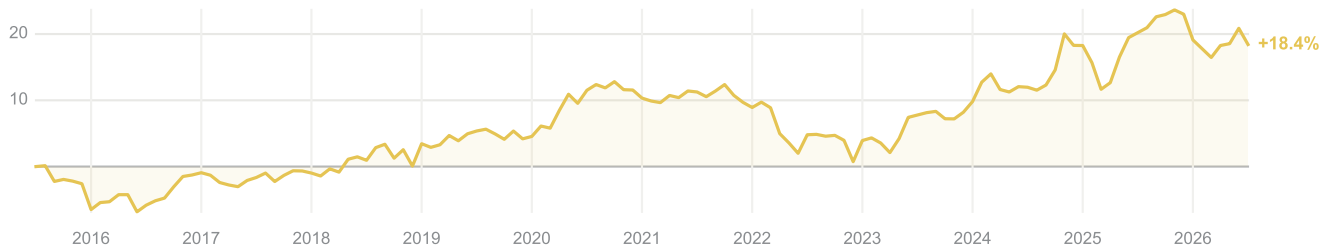
CAGR +3.1%



223 ● **Payout Ratio**

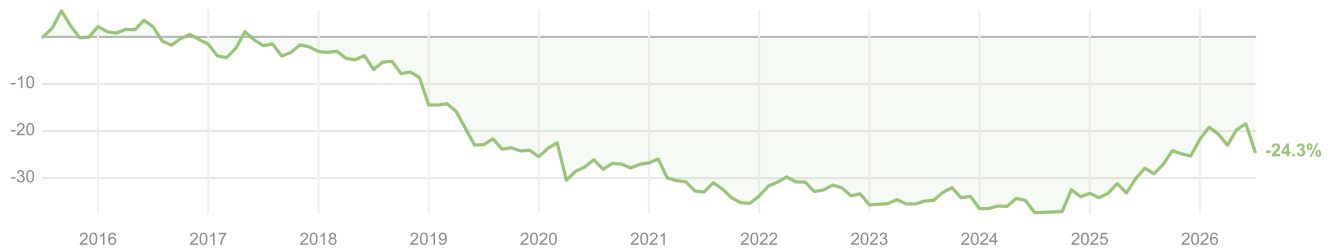
Capital Efficiency

CAGR +3.0%

224 ● **10 Day MACD Trend**

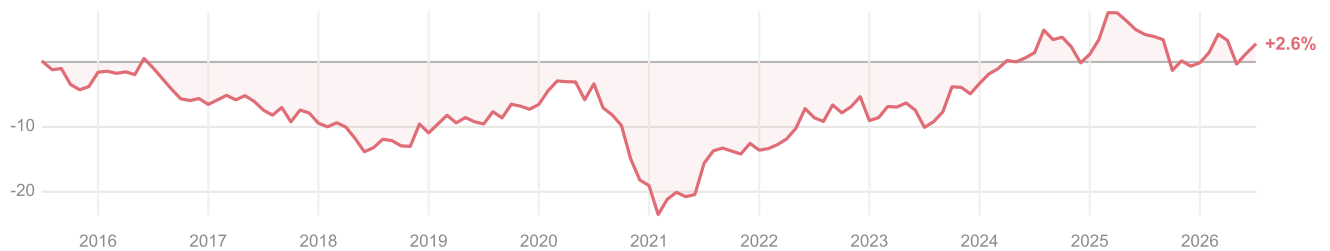
Price Momentum

CAGR +3.0%

225 ● **Change of Sales to Change of EPS**

Earnings Quality

CAGR +3.0%

226 ● **6M Chg in 18M CAPM Alpha**

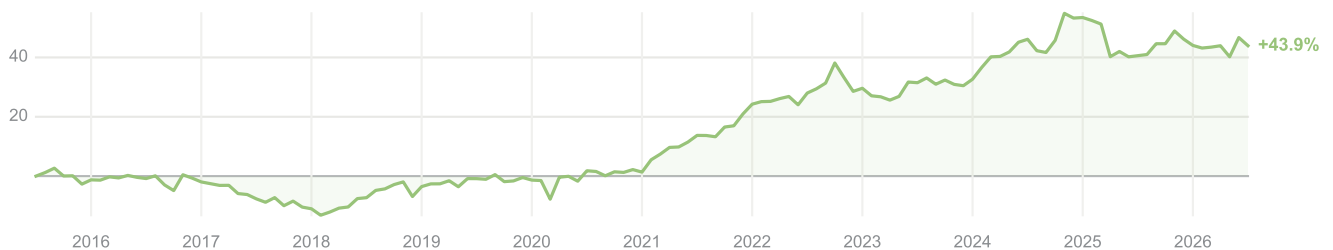
Price Momentum

CAGR +3.0%

227 ● **Investment Duration**

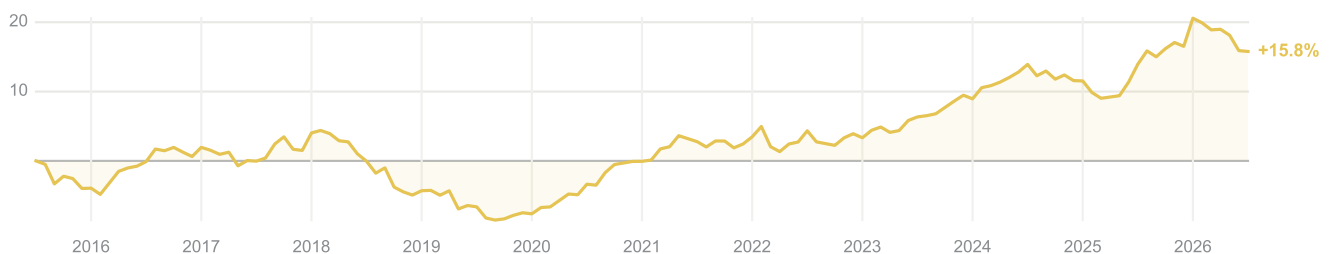
Price Momentum

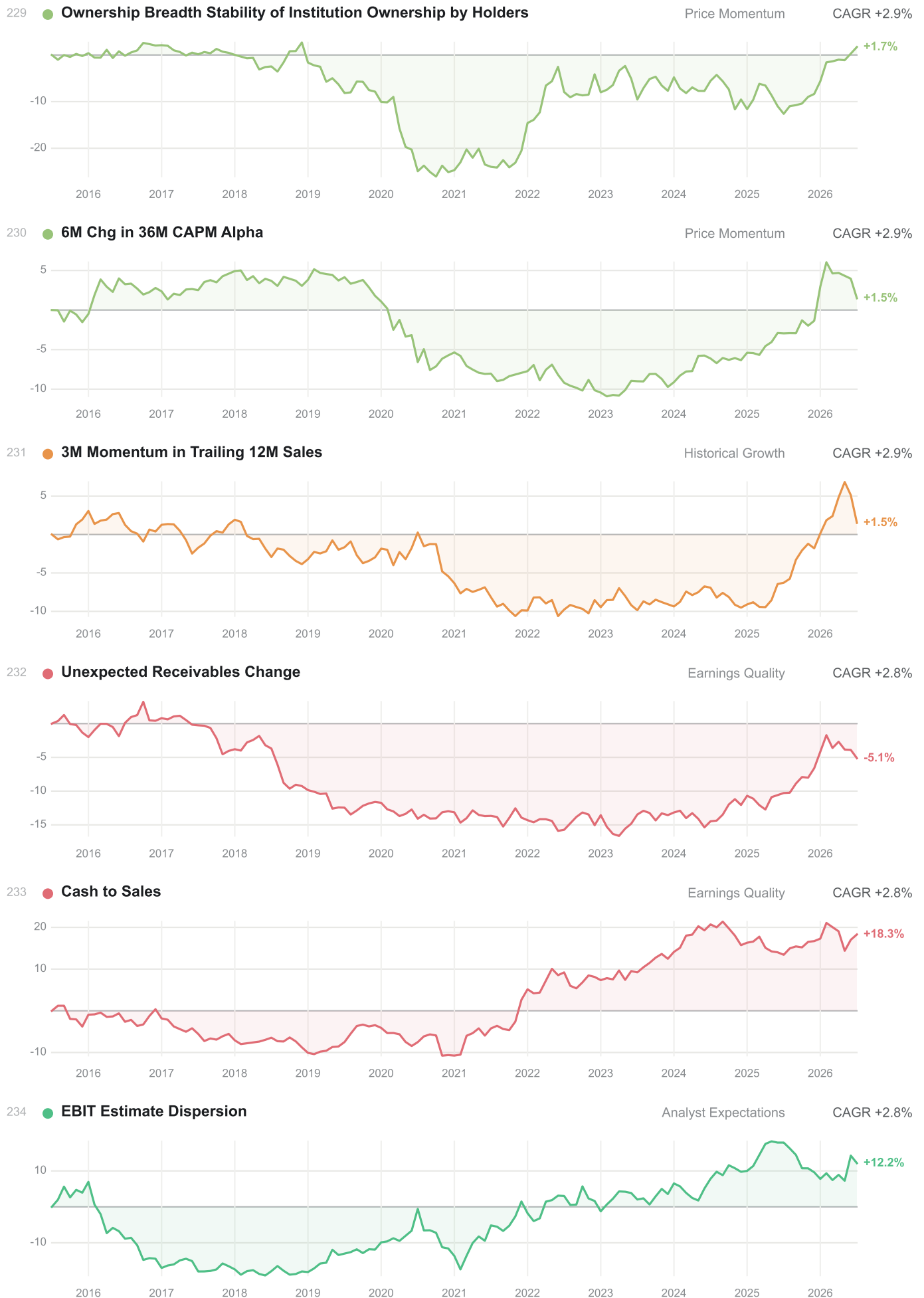
CAGR +2.9%

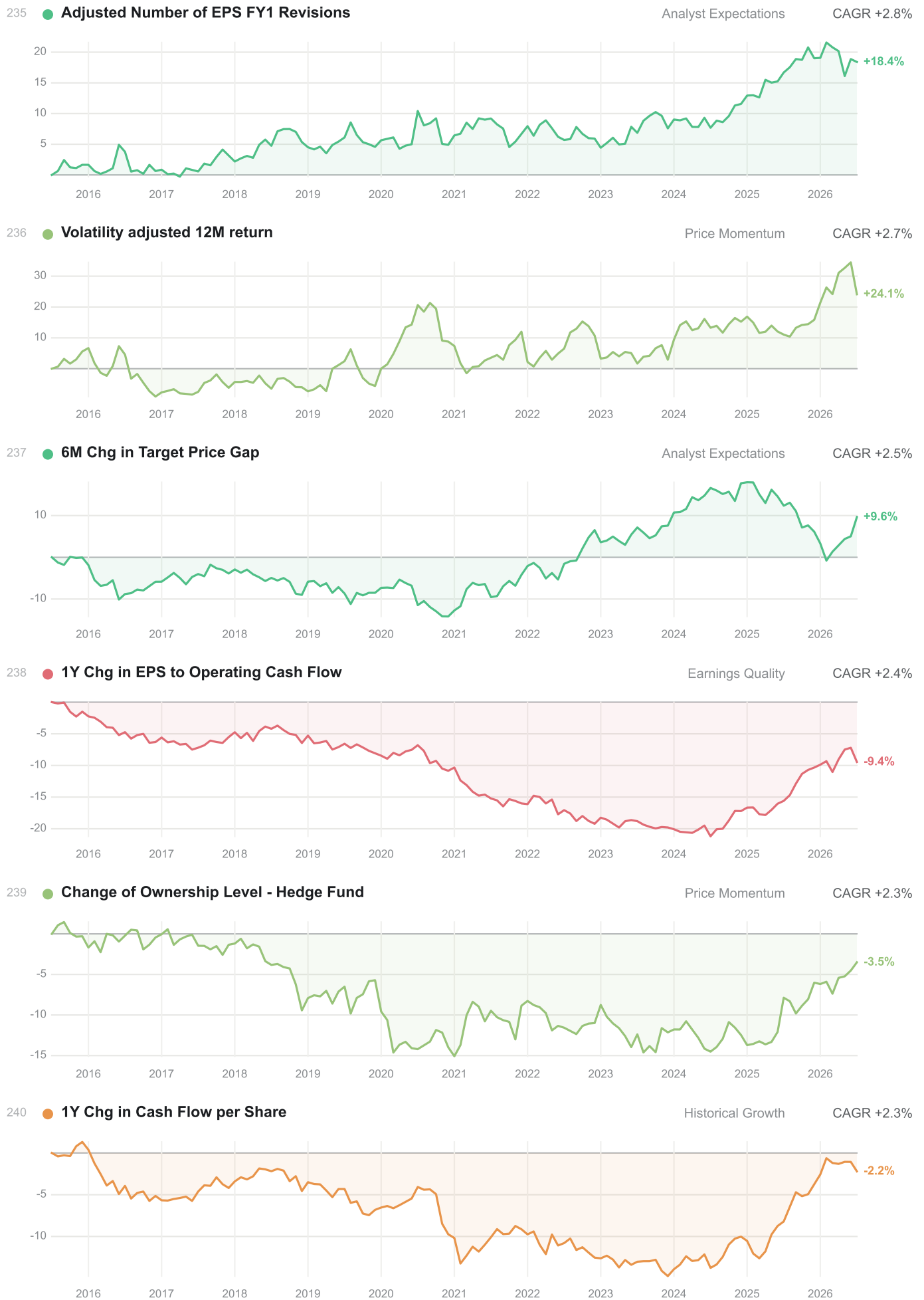
228 ● **Working Capital Turnover Ratio**

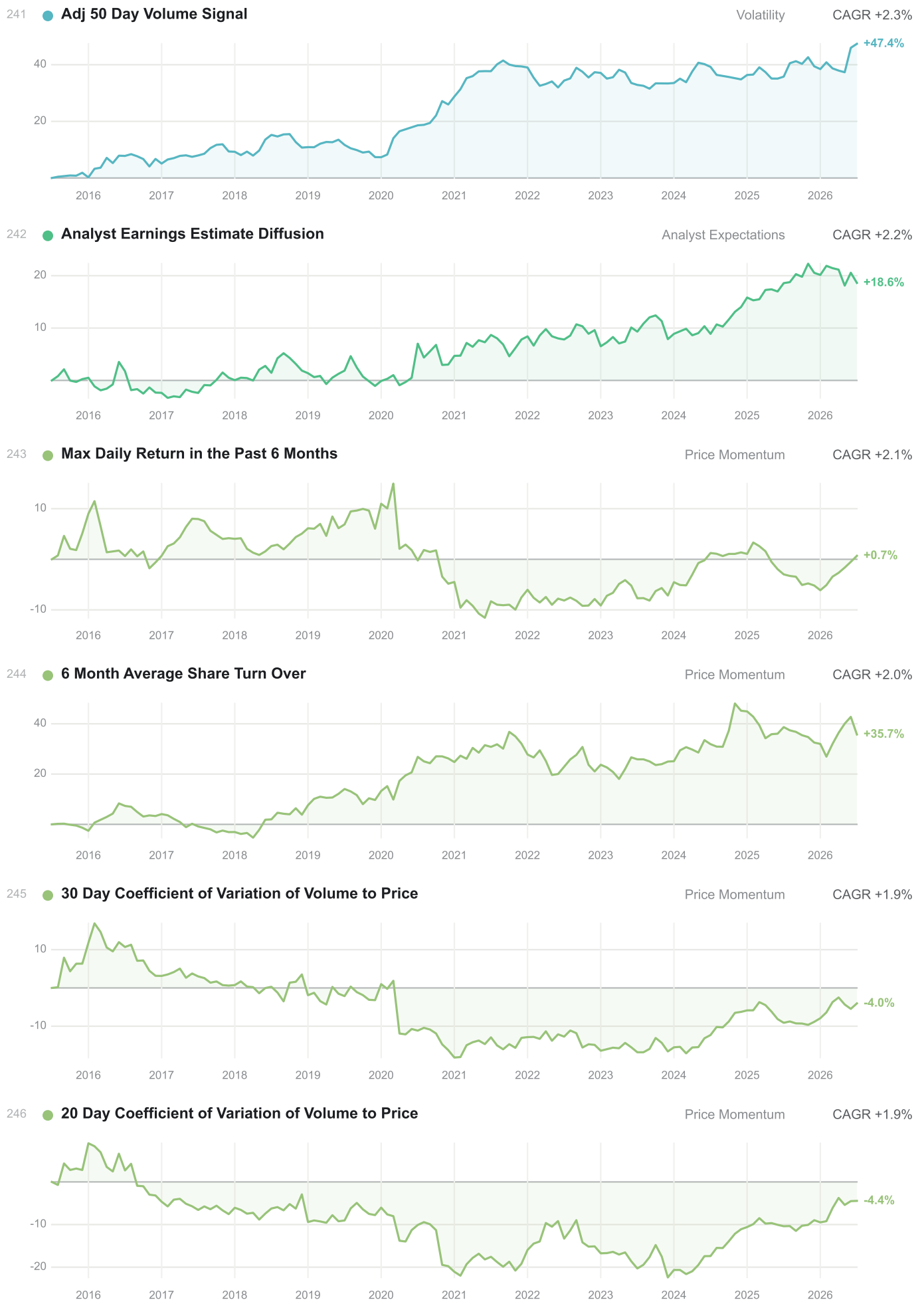
Capital Efficiency

CAGR +2.9%









247 ● **Year over Year Change of Inventory to Assets**

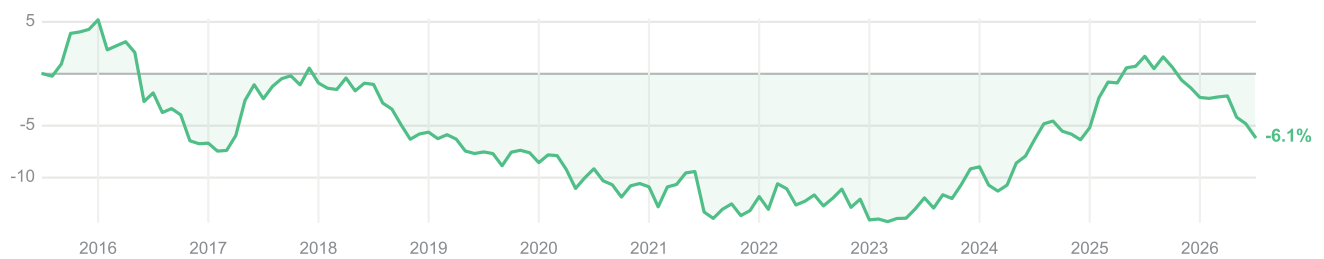
Earnings Quality

CAGR +1.8%

248 ● **Buy-to-Sell Recommendation Ratio Less 3M SMA**

Analyst Expectations

CAGR +1.8%

249 ● **Closing Price to 52 Week High**

Price Momentum

CAGR +1.8%

250 ● **Institution Ownership Turnover**

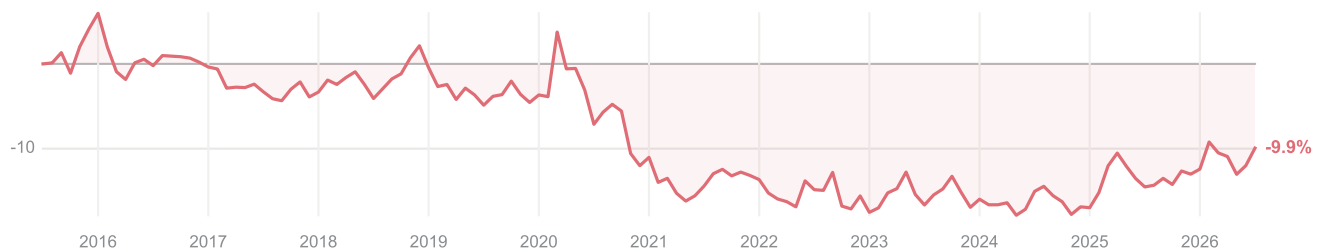
Price Momentum

CAGR +1.7%

251 ● **ROE 20 Qtr Standard Deviation**

Earnings Quality

CAGR +1.5%

252 ● **6M Chg in 12M CAPM Alpha**

Price Momentum

CAGR +1.3%



253 ● **Sales Acceleration**

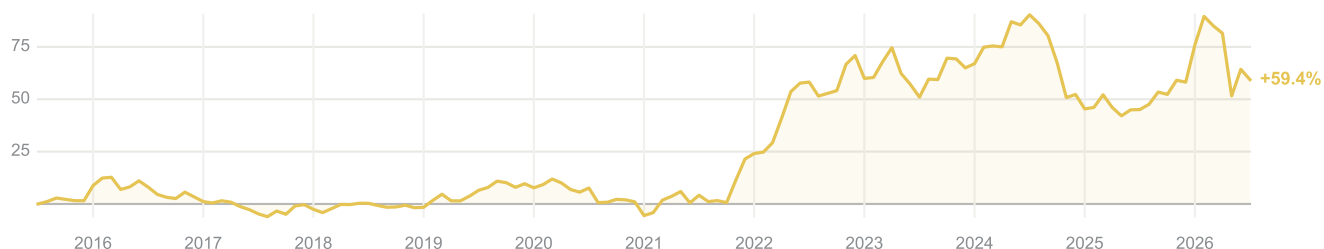
Historical Growth

CAGR +1.3%

254 ● **Normalized ROA**

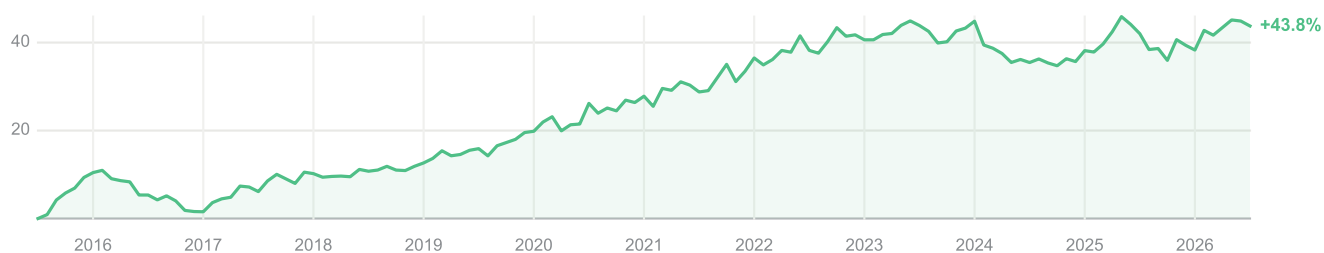
Capital Efficiency

CAGR +1.2%

255 ● **6M Avg Chg 1M Recommendation**

Analyst Expectations

CAGR +1.1%

256 ● **14-Day Relative Strength Index**

Price Momentum

CAGR +0.9%

257 ● **Change in TTM Depr. to Capex**

Earnings Quality

CAGR +0.8%

258 ● **Buy-to-Sell Recommendation Ratio Less 3M EMA**

Analyst Expectations

CAGR +0.6%



